

Gender homophily in Italian music collaborations

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Collaboration patterns among Italian musicians

Gender homophily and musical genre on Discogs

This report was generated automatically at an earlier stage of the analysis and has not been revised since. Its interpretation of the time series (a reversal of sign) was later withdrawn; the current results are in docs/06-risultati.md and in the manuscript. The figures below are recomputed from the current data files, but the surrounding text may no longer match them.

Analysis run on 26 September 2026. Single source: a local Discogs dump (PostgreSQL).

Summary for the busy reader

This study reconstructs **who recorded with whom** among the Italian musicians listed on Discogs and asks whether people's gender structures those collaborations. The short answer is yes, but not in the way one would expect. Homophily exists and is statistically very solid, but it is much weaker than the separation by musical genre; it is **asymmetric** (it is women who cluster together, not men, the opposite of what the current literature reports); and it did not strengthen after 2000: what strengthened is the network's tendency to **close into triangles**, which produces the same apparent effect for a different reason.

The numbers

Italian musicians identified	87,229
of whom with determined gender	65,093 (74.6%)
Share of women among those with determined gender	14.8% (a ratio of 5.77 men for every woman)
Credits analyzed	3,682,616, of which 2,360,057 (64.1%) resolved to the individual track
Collaboration network	70,712 connected artists, 586,040 ties
Giant component	95.4% of connected artists
Gender assortativity	r = 0.0432 (95% CI 0.0399–0.0465)
Musical-genre assortativity	r = 0.4970
Gender homophily pre-2000 → post-2000	0.0406 → 0.0693 (difference not significant in the ERGM)
ERGM homophily, women vs. men (median across subnetworks)	0.679 vs. 0.074 in log-odds
ERGM	8 subnetworks, 3 models each, 8 converged , gwesp included

The five things to know

1. **Italian recorded music is a men's world, and it has remained one.** A share of 14.8% women among artists with determined gender means 5.8 men for every woman. The share does not grow monotonically over time: it starts at 17.8% for those who debuted in the 1960s, falls to 13.2% in the 1980s and rises again to 21.4% for those who debuted from 2020 onward.
2. **Musical genre separates far more than gender does.** Assortativity by musical genre (0.497) is about 12 times that by gender (0.043). Jazz musicians record with jazz musicians far more systematically than men record with men.
3. **Gender homophily is small but real.** $r = 0.0432$ looks small, but the degree-preserving null model gives 0.0001 and the confidence interval (0.0399–0.0465) lies entirely above zero. It is not a composition effect; it is structure.
4. **It is women who cluster together, not men.** This is the result that most clearly contradicts expectations. In the ERGM, which holds activity, cohort, musical genre and triadic closure constant, the female homophily coefficient is positive and large in **all eight** estimated subnetworks; the male coefficient is close to zero, and in Rock it is even negative. Where a group is an overwhelming majority it has no need to seek itself out; where it is rare, it clusters.
5. **After 2000 it is not homophily that increases, but closure into triangles.** Descriptively, assortativity rises from 0.0406 to 0.0693. But in the ERGM the difference between the two periods in the gender terms is **not significant** ($p = 0.88$ for women, 0.60 for men), while the triadic closure term grows from 2.22 to

2.89 with $p < 0.0001$. The criterion by which a collaborator is chosen has not changed: the shape of the network has.

6. **The “Smurfette” pattern is not observed.** Holding releases, cohort and musical genre constant, being a woman does not shift network position on any of the three centrality measures used (section 6.2). The inequality is large, but it lies in **access** and in the volume of activity, not in the position of those who managed to get in.

Measurement note. All gender assortativities reported as main results are computed only on edges where **both** artists have a determined gender (80.3% of edges). Including *unknown* as a fourth category systematically inflates the index, because poorly documented artists collaborate with each other more than chance would predict, for reasons of data coverage. The two measures are compared in a table in section 5.1.

How far to trust these results

Gender does not appear in any source: it is **inferred**. A determination is reached for 74.6% of the population, with 5,203 cases anchored to Wikidata through the Discogs identifier (exact join, no homonymy) and the rest inferred from first names. The manual validation sample of 200 cases and the tool for measuring the error are ready in `data/validation_sample.csv`; until the sample is coded by hand, the figures above should be read as estimates with an error that has not yet been quantified. The Monte Carlo analysis in section 8.1 nonetheless shows that homophily remains positive **in every imputation scenario**, including the one built specifically to minimize it.

1. Data sources: what is there, what is missing, what had to be built

1.1 The source

The analysis uses a **single source**: a local copy of the Discogs dump in PostgreSQL (database `discogs`, 248,153,636 rows in the tables used). The initial design also included a second database, iTunes, and a bridge table between the two; at the client's request iTunes was **excluded**. The methodological consequence is clear-cut and must be stated: there is no independent cross-check of the artist records, and the robustness axis "union of sources vs. Discogs only" no longer applies. In its place, two alternative axes were introduced (the Italian-share threshold and the use or non-use of track-level credits), discussed in section 8.2.

All database sessions ran with `default_transaction_read_only = on`. In PostgreSQL this setting also forbids temporary tables, so the extraction was rewritten to create **no** objects on the server: lists of identifiers computed in Python are sent back to the database as `int[]` literals. No write of any kind touched the source.

1.2 Three gaps that changed the design

Exploration found three gaps in the dump that forced deviations from the original plan. They need to be stated clearly because they limit what can be concluded.

release_label is empty (0 rows). There is no link between releases and record labels. The planned heuristic for identifying Italian musicians through *Italian labels* therefore cannot be applied, and with it goes any analysis by record company. Italian identity therefore rests on the country of release alone.

release_genre and release_style are empty (0 rows). Musical genre is available only through the *master* (`master_genre`), and masters cover about 59% of releases. This is why 26.9% of the population has no musical genre assigned.

The "Various Artists" entity is effectively absent. In the whole of `release_artist` (over 92 million rows) there are **328** main credits attributed to an artist named "Various*". In this dump compilations do not use the usual placeholder: they list the artists directly. The `exclude_various` filter is therefore almost inert (3 releases caught), and compilations have to be recognized by a different criterion, the presence of several distinct main artists, which is the `exclude_compilations` filter used as a robustness axis.

1.3 Who counts as an “Italian musician”

Without label data and without a nationality field, Italian identity was defined by the **share of Italian releases** in each artist’s career:

An artist enters the population if they have at least 2 releases with country = 'Italy', at least 3 releases in total, and if the Italian ones make up at least 50% of the total.

The share rule is the part that does the work. The absolute count alone produces a list dominated by Beethoven, Mozart, Bach, Chopin and Karajan: the classical repertoire is reissued in Italy in large quantities, and counting only “how many Italian releases” mistakes the catalog for the biography. The share excludes all of them, because for each of them the Italian editions are a tiny fraction of a worldwide catalog.

The validity check consisted of inspecting the top twenty-five artists by volume: Mina, Lucio Battisti, Vasco Rossi, Fabrizio De André, Franco Battiato, Lucio Dalla, Mogol, Renato Zero, Francesco De Gregori, Domenico Modugno, Claudio Villa, plus a group of Italian producers and engineers who are genuinely active (Antonio Baglio, Giovanni Versari, Vincenzo Tempera). No evident false positives.

Two structural limits remain that no threshold can remove:

- the rule measures **where people release**, not **where they come from**. A foreign musician who has worked almost exclusively for the Italian market enters the population; an Italian emigrant who releases mostly abroad drops out. The 50% threshold keeps the first error low at the cost of increasing the second;
- Discogs is not a census. It overrepresents vinyl, collecting and electronic music, and underrepresents music that was never released on a cataloged physical format. The composition by musical genre in section 3.2 should be read as the composition of *the Discogs catalog*, not of Italian music.

Effect of the threshold (measured): 254,286 artists have at least two Italian releases; applying the share and the minimum, the population falls to **87,229**. Raising the share to 0.60 and 0.70 gives 78,448 and 66,113 artists: section 8.2 shows that the conclusions do not change.

2. Gender: how it was inferred

2.1 The problem

None of the available sources records people's gender. It has to be inferred, and the inference must be documented in full, because it is the most fragile point of the whole study: every conclusion about gender homophily rests on a label that nobody declared.

The cascade proceeds from the strongest signal to the weakest, and each artist carries **which level** their label comes from and **with what confidence**.

2.2 Level 1: Wikidata linked to the Discogs identifier

Wikidata exposes property P1953, *Discogs artist ID*. This allows an **exact join on the identifier**, not on the name: no homonymy, no approximate matching. All entities with P1953 and P21 (sex or gender) were collected: **202,478 distinct Discogs identifiers**, with 74 ambiguous cases discarded and **zero blocks lost** in a recursive pagination by identifier prefix.

Of these, **5,203 fall within our population** (6.0%). This is low coverage in absolute terms, and the reason is obvious: Wikidata describes notable people, while the Discogs population consists largely of session musicians, arrangers, sound engineers and producers who have no encyclopedia entry. But these are 5,203 **certain** labels, and the next level rests on them.

2.3 Level 2: first names, learned from the data rather than taken from a list

The design called for the ISTAT list of first names. In this environment it was not available as a downloadable dataset; the fallback adopted is better for the purpose, not worse.

The same 202,478 Discogs identifiers labeled by Wikidata were joined back to the **names** in the database: this yields **199,812 name→gender pairs for musicians**, a domain-specific corpus of first names, much larger than the Italians alone and much more relevant than a generic civil-registry list.

Two dictionaries are derived from it, and the order in which they are consulted is the most important methodological choice in this section:

1. **Italian dictionary** (320 names), built only on the Italian artists labeled by Wikidata;
2. **gender-guesser with the Italian lookup;**

3. **global dictionary** (2,204 names), but **only for names that the Italian lookup does not recognize**;
4. global gender-guesser, as a last resort and with reduced confidence.

The constraint in point 3 is not pedantry. Andrea, Simone, Nicola, Daniele, Michele and Gabriele are male names in Italy and female names in English-dominated dictionaries: using the global dictionary without that filter would flip the gender of some of the most common Italian male first names, a systematic rather than random error, concentrated precisely on the most frequent names. Checked against the data: the two dictionaries agree on all 241 names they have in common, which indicates that the filter is actually keeping the two domains apart rather than masking a conflict.

2.4 Level 3: groups are read from their members

A band name says nothing about the gender of its members. For groups, therefore, the membership is examined (`group_member`): if the members of known gender include both genders the group is **mixed**, if they are all of the same gender the group inherits it, and if fewer than two are known it remains unknown. **0 of 0 groups** were resolved, of which 0 are mixed.

2.5 Outcome of the cascade

source	artists	share
onomastico_prior_it	44,618	51.2%
none	22,136	25.4%
onomastico_prior_globale	6,217	7.1%
wikidata_p1953	5,203	6.0%
onomastico_gg_it_male	3,196	3.7%
onomastico_gg_it_female	2,262	2.6%
wikidata_name	1,500	1.7%
onomastico_gg_globale_male	1,162	1.3%
onomastico_gg_globale_female	863	1.0%
onomastico_gg_it_mostly_male	44	0.1%
onomastico_gg_it_mostly_female	28	0.0%

Table: Source of the gender label for each artist. The `onomastico_prior_it` row carries most of the load on its own: it is the dictionary built on the 5,203 certain Italians, and 320 first names are enough to cover almost half of the population, because Italian first names are highly concentrated. `none` and `group_unresolved` are the two faces of not knowing: artists whose name is not a recognizable personal name (acronyms, pseudonyms, projects) and groups for which not enough members are known.

Final outcome: 14.8% women among artists with determined gender, that is, 5.77 men for every woman, with 25.4% of the population remaining undetermined.

2.6 The missing piece: manual validation

data/validation_sample.csv has been generated: **200 artists** drawn with stratification by gender, confidence band and label source, with an empty human_gender column to be filled in by hand. The script src/score_validation.py computes precision, recall, F1 and the confusion matrix by cascade level as soon as the file is filled in.

The sample uses **half proportional and half uniform allocation** across strata: the proportional part makes it possible to estimate overall accuracy without reweighting, and the uniform part guarantees enough cases even in the rare levels of the cascade. Within each stratum, distinct first names are favored, because otherwise small strata fill up with people who share a name and the validation would measure accuracy on one name rather than on a level.

On two levels that remedy is not enough, and this should be said:

onomastico_gg_mostly_female contains 28 artists who share **a single** first name (Mary), and onomastico_gg_mostly_male contains 44 with two (Toni, Leonida). On those two levels the validation will be able to say whether those names are classified correctly, not whether the level works in general. Together they account for 72 of 87,229 artists, so this does not affect the conclusions, but the resulting accuracy figure should not be read as generalizable.

This step **has not been carried out**: it requires human judgment. Until it is completed, the error of the gender inference is bounded only by the Monte Carlo analysis in section 8.1, which, however, measures the effect of *uncertainty about the unknowns*, not that of *errors among the knowns*. This is the most serious limitation of the study.

3. The network: how two musicians become connected

3.1 Edge weight, and why not all credits count the same

A Discogs credit can say three very different things. It can say “*this person plays bass on track B2*”; it can say “*this person is the artist of the record*”; it can say “*this person appears in the credits of the record*”, without specifying where. Treating them as equivalent would mean giving a shared sleeve credit the same value as a documented session.

Each credit therefore receives a **specificity**, and the weight of the tie between two artists uses it:

scope	specificity	what it means
track	1.00	credit resolved to a specific track
main	0.70	main artist of the release
umbrella	0.30	secondary credit, with no indication of tracks

$$w(u,v) = w_t \cdot |\text{shared tracks}| + w_r \cdot \sum_R s_u(R) s_v(R) \quad w(u,v) = w_t \cdot |\text{shared tracks}| + w_r \cdot \sum_R s_u(R) s_v(R)$$

The first term rewards collaboration **documented on the same track**; the second retains co-presence on the same release, scaled by the specificity of both credits. Two session musicians credited on the same track weigh much more than two names that appear generically on the same record.

Track-level credits come from two routes. The first is `release_track_artist`, which carries a global track identifier: **1,932,237 credits**. The second is the `tracks` field of `release_artist`, which gives positions in text form (A1, 1 to 3, 4, 6, 12) and has to be **resolved**: positions are converted into actual tracks through `release_track`. Of 442,050 positional credits, 427,820 (96.8%) were resolved; the rest use free-form expressions (“*all tracks except 1, 13 and 14*”) and were **downgraded to umbrella** rather than forced into an interpretation.

Total: **2,360,057 of 3,682,616 credits (64.1%) are resolved at the level of the individual track.**

3.2 Filters and subnetworks

Releases with more than 8 credited artists are discarded: they are compilations and box sets, where co-presence does not indicate collaboration and the number of pairs grows quadratically. The filter reduces the credits from 3,682,616 to 586,040.

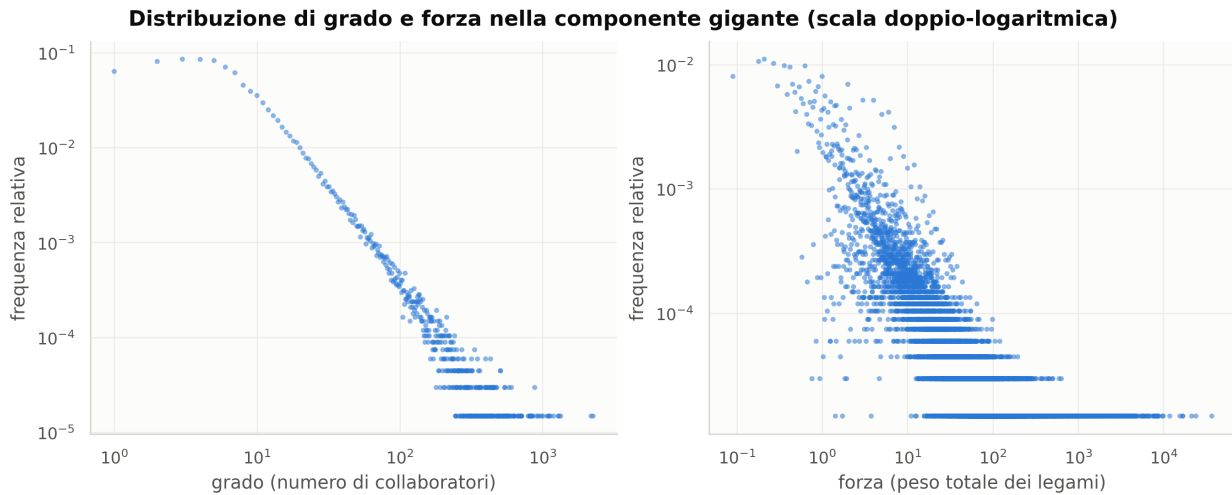
The role subnetworks separate two different trades: **creative** (production, writing, arrangement, composition) and **performance** (vocals, instruments, conducting, featuring). A main artist without an explicit role is treated as a performer, because that is what being the artist of a record means.

network	total nodes	nodes with edges	edges	density	mean degree	median degree	max degree	components	giant component	giant component size
all	87,229	70,712	586,040	0.000234	16.575404	7.000000	2,258	1,204	67,443	0.954
creative	87,229	30,440	199,832	0.000431	13.129566	4.000000	1,348	1,530	26,538	0.871
performance	87,229	49,152	211,751	0.000175	8.616170	4.000000	598	2,493	42,013	0.856

Table: Descriptive statistics of the three collaboration networks

Full data in `tables/t2_rete_descriptive.csv` and `tables/t2_rete_descriptive.tex`.

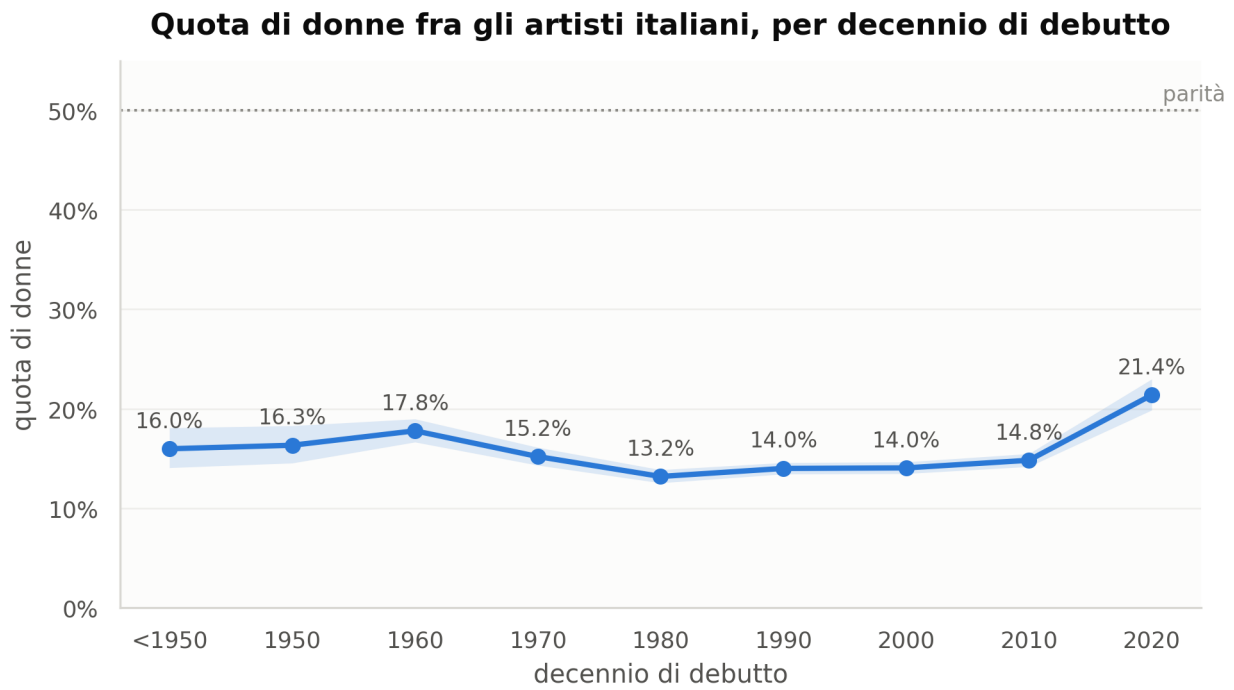
The overall network is **sparse and highly connected**: density on the order of 0.000234, but a giant component that absorbs 95.4% of the connected artists. This is the typical signature of a professional world in which almost nobody works in isolation and almost nobody works with everyone. The creative network is smaller and denser than the performance network: producers and songwriters form a tighter and more interwoven core than performers, which matters for the reading of section 5.3, where the two roles show different levels of homophily.



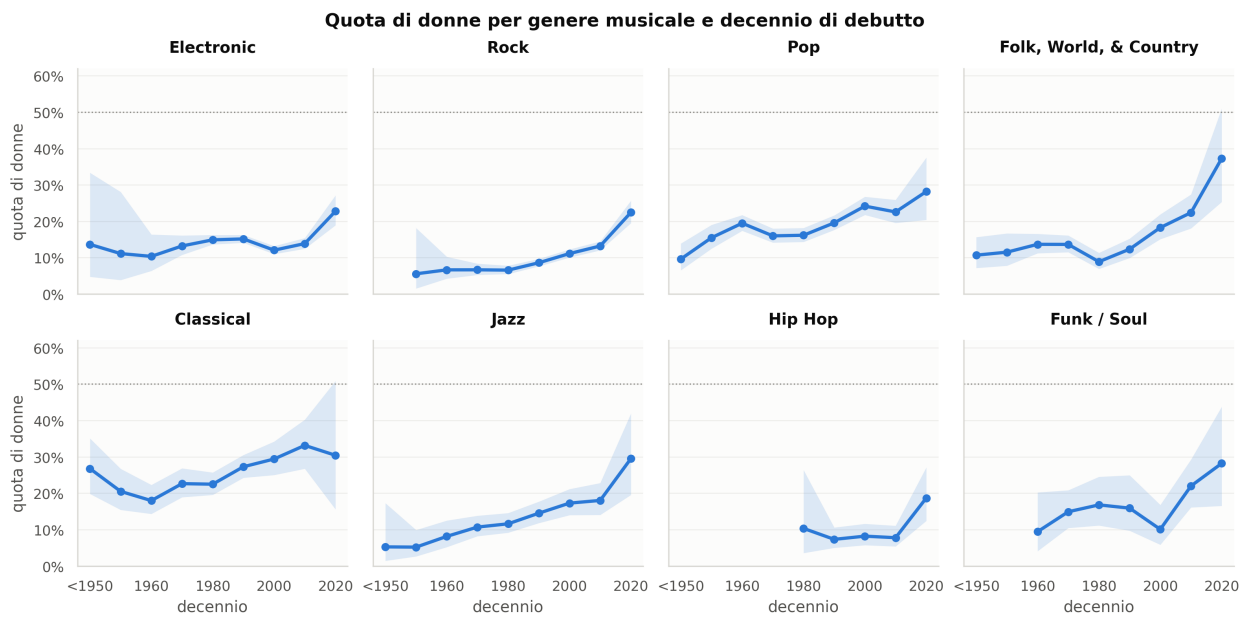
Degree and strength distributions. On a log-log scale both distributions decline along an almost straight line over several orders of magnitude: the vast majority of artists have very few collaborators, while a thin minority have hundreds. Strength, which sums the weights and therefore counts how many times people collaborated and how specific the credits were, has an even longer tail than degree: the major collaborators do not just have many partners, they have much more intense relationships. This is the structural background against which section 6.1 should be read: in a network this unequal, the question ‘are women at the center?’ must always be asked holding activity constant, because the number of releases alone already explains much of centrality.

The analyses that follow run on the **giant component**, because centrality measures and distances are not defined across separate components. The excluded share is 4.6% of the connected artists.

4. How many women there are, and in which music (RQ1)



The share of women by decade of debut. The line is not a story of progress. It starts at 17.8% for those who debuted in the 1960s, falls to 13.2% in the 1980s (the lowest point in the series) and rises only recently, to 21.4% for those who debuted from 2020 onward. The decline of the 1970s and 1980s calls for caution before it is read as a real setback: it coincides with the massive expansion of the Discogs catalog for those years, that is, with the mass entry of technical and production credits (trades that are almost entirely male), which dilute a share computed over all credits rather than over performers only. The recent rise, by contrast, is consistent in both magnitude and direction with what is observed in other music catalogs. In every decade, however, the confidence band remains very far from parity.



Share of women by musical genre and decade. The panels show that there is no single trajectory of gender representation: there are musical genres with different histories. The starting level matters more than the slope: a genre that starts low tends to stay low across the decades, which is exactly the signature of a segregation that reproduces itself through recruitment rather than dissolving over time.

musical genre	debut decade	artists	gender known	women	share of women	ci_lo	ci_hi	men per woman
Blues	1940	1	1.000	0.000	0.000	0.000	0.793	n/a
Blues	1950	3	0.000	0.000	n/a	n/a	n/a	n/a
Blues	1960	17	13.000	1.000	0.077	0.014	0.333	12.000
Blues	1970	61	54.000	9.000	0.167	0.090	0.287	5.000
Blues	1980	105	98.000	15.000	0.153	0.095	0.237	5.533
Blues	1990	129	106.000	15.000	0.142	0.088	0.220	6.067
Blues	2000	137	121.000	21.000	0.174	0.116	0.251	4.762
Blues	2010	141	114.000	22.000	0.193	0.131	0.275	4.182
Blues	2020	22	17.000	4.000	0.235	0.096	0.473	3.250
Brass & Military	1940	24	14.000	0.000	0.000	0.000	0.215	n/a
Brass & Military	1950	10	6.000	0.000	0.000	0.000	0.390	n/a
Brass & Military	1960	38	18.000	2.000	0.111	0.031	0.328	8.000
Brass & Military	1970	13	6.000	0.000	0.000	0.000	0.390	n/a
Brass & Military	1980	9	5.000	0.000	0.000	0.000	0.434	n/a
Brass & Military	1990	12	9.000	0.000	0.000	0.000	0.299	n/a
Brass & Military	2000	1	1.000	0.000	0.000	0.000	0.793	n/a
Brass & Military	2010	1	0.000	0.000	n/a	n/a	n/a	n/a
Children's	1940	33	23.000	10.000	0.435	0.256	0.632	1.300
Children's	1950	79	72.000	24.000	0.333	0.235	0.448	2.000
Children's	1960	208	173.000	66.000	0.382	0.312	0.456	1.621
Children's	1970	170	131.000	70.000	0.534	0.449	0.618	0.871
Children's	1980	85	53.000	24.000	0.453	0.327	0.585	1.208
Children's	1990	60	45.000	16.000	0.356	0.232	0.502	1.812
Children's	2000	27	23.000	7.000	0.304	0.156	0.509	2.286
Children's	2010	13	9.000	2.000	0.222	0.063	0.547	3.500

Table: Share of women by musical genre and decade of debut, with 95% Wilson intervals

First 25 of 141 rows shown; the full table is in `tables/t3_quota_donne_genere_decennio.csv` and `tables/t3_quota_donne_genere_decennio.tex`.

4.1 Comparison with the patterns reported in the literature

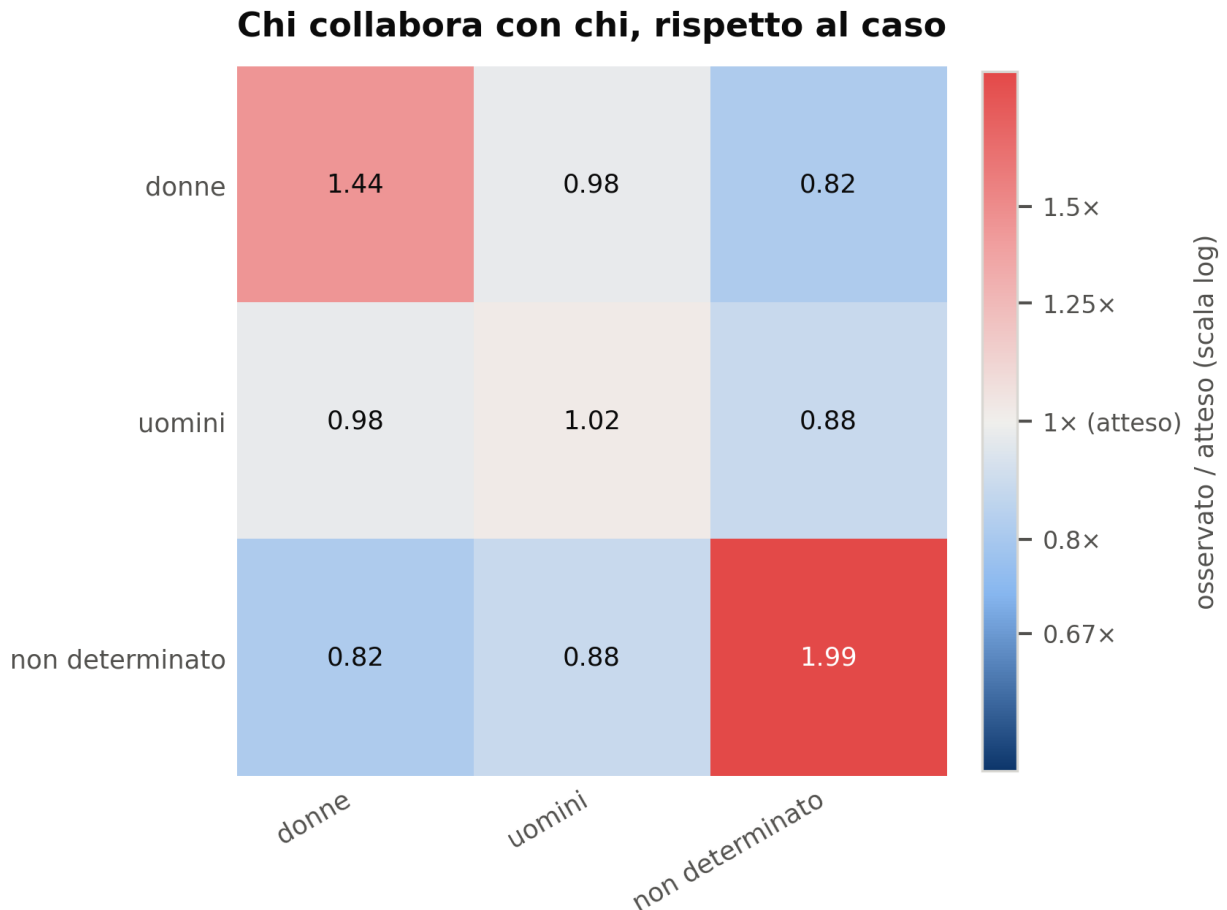
The usual benchmark for hip hop is a ratio of about **4 men for every woman**. In the Italian data the ratio is **10.4 to 1** (8.8% women): markedly **more unbalanced** than the international benchmark. There are two readings, not mutually exclusive: the Italian hip hop scene recorded in Discogs is smaller and more recent, and therefore more exposed to the fact that production roles, where women are rarer, weigh relatively more; and the count here includes all credits, not only lead performers, which lowers the share compared with chart-based statistics.

At the opposite end, **Classical** (24.7%) and **Children's** (41.3%) are the genres with the highest female presence, the latter above 41%, the only genre in the entire corpus in which women approach half. The gap between Children's and Hip Hop, with the same population and the same method, is more than 33 percentage points: musical genre is the strongest predictor of female presence in this entire study.

Rock (10.5%) and Electronic (14.2%), which together make up the largest part of the population, are both below the overall average. Given their numerical weight, it is these two scenes that determine the overall share.

5. Homophily: who records with whom (RQ2)

5.1 The overall picture



Who collaborates with whom, relative to chance. Each cell is the ratio between the observed ties and those expected under a null model that preserves exactly each artist's degree and the composition of the population: the only thing randomized is *who is with whom*. A value of 1 means 'as by chance', above 1 means more than expected. A diagonal above 1 and off-diagonal cells below 1 are the operational definition of homophily. The unknown-unknown cell also departs from 1: this is not a social fact but a matter of data coverage. Artists about whom we know nothing tend to be together because they share the same characteristics that make them poorly documented (few credits, minor roles, marginal periods). This is precisely why the ERGM in section 7 excludes unknown nodes instead of treating them as a category.

attribute	r observed	r weighted	r null (mean)	r null (SD)	z
gender	0.0810	0.1254	-0.0000	0.0010	81.4605
musical_genre	0.4970	0.6675	-0.0001	0.0006	790.6131

Table: Assortativity, observed and under the null model

Full data in `tables/t3_assortativita_globale.csv` and `tables/t3_assortativita_globale.tex`.

subnetwork	stratum	attr.	categories	edges	edge share	r	ci_lo	ci_hi
all	all	gender	determined only	470,690	0.8032	0.0432	0.0399	0.0465
all	all	gender	all categories	586,040	1.0000	0.0810	0.0788	0.0834
all	all	musical_genre	determined only	503,971	0.8600	0.5622	0.5606	0.5638
all	all	musical_genre	all categories	586,040	1.0000	0.4970	0.4955	0.4984
all	pre2000	gender	determined only	333,067	0.8444	0.0406	0.0367	0.0443
all	pre2000	gender	all categories	394,423	1.0000	0.0576	0.0548	0.0605
all	pre2000	musical_genre	determined only	362,127	0.9181	0.5103	0.5082	0.5124
all	pre2000	musical_genre	all categories	394,423	1.0000	0.4704	0.4685	0.4721
all	post2000	gender	determined only	61,015	0.6563	0.0693	0.0598	0.0788
all	post2000	gender	all categories	92,968	1.0000	0.1479	0.1416	0.1533
all	post2000	musical_genre	determined only	64,388	0.6926	0.7101	0.7062	0.7141
all	post2000	musical_genre	all categories	92,968	1.0000	0.5445	0.5407	0.5482
creative	all	gender	determined only	179,908	0.9003	0.0224	0.0172	0.0279
creative	all	gender	all categories	199,832	1.0000	0.0703	0.0658	0.0750
creative	all	musical_genre	determined only	184,147	0.9215	0.5770	0.5743	0.5799
creative	all	musical_genre	all categories	199,832	1.0000	0.5351	0.5323	0.5377
creative	pre2000	gender	determined only	146,391	0.9325	0.0224	0.0165	0.0282
creative	pre2000	gender	all categories	156,982	1.0000	0.0269	0.0224	0.0316
creative	pre2000	musical_genre	determined only	149,499	0.9523	0.5209	0.5177	0.5243
creative	pre2000	musical_genre	all categories	156,982	1.0000	0.4945	0.4914	0.4977
creative	post2000	gender	determined only	13,668	0.7079	0.0444	0.0227	0.0655

subnetwork	stratum	attr.	categories	edges	edge share	r	ci_lo	ci_hi
creative	post2000	gender	all categories	19,308	1.0000	0.1616	0.1478	0.1753
creative	post2000	musical_genre	determined only	14,659	0.7592	0.7868	0.7788	0.7953
creative	post2000	musical_genre	all categories	19,308	1.0000	0.6235	0.6145	0.6320
performance	all	gender	determined only	160,881	0.7598	0.1054	0.0996	0.1115
performance	all	gender	all categories	211,751	1.0000	0.1704	0.1667	0.1741
performance	all	musical_genre	determined only	171,934	0.8120	0.6153	0.6125	0.6177
performance	all	musical_genre	all categories	211,751	1.0000	0.5305	0.5279	0.5327
performance	pre2000	gender	determined only	95,250	0.8164	0.1106	0.1028	0.1180
performance	pre2000	gender	all categories	116,673	1.0000	0.1733	0.1679	0.1785
performance	pre2000	musical_genre	determined only	103,792	0.8896	0.5612	0.5582	0.5647
performance	pre2000	musical_genre	all categories	116,673	1.0000	0.5132	0.5101	0.5164
performance	post2000	gender	determined only	31,344	0.6306	0.1050	0.0913	0.1180
performance	post2000	gender	all categories	49,702	1.0000	0.1985	0.1909	0.2066
performance	post2000	musical_genre	determined only	33,316	0.6703	0.7482	0.7428	0.7533
performance	post2000	musical_genre	all categories	49,702	1.0000	0.5737	0.5685	0.5787

Table: Assortativity computed only on nodes with a determined attribute, compared with the computation that treats ‘undetermined’ as a category, for gender and for musical genre

Full data in `tables/t3_assortativita_MF_vs_tutte.csv` and `tables/t3_assortativita_MF_vs_tutte.tex`.

A measurement issue that changes the numbers

The two tables should be read together, because the second corrects the first.

Treating unknown as a fourth category on a par with M and F inflates assortativity: the unknown-unknown cell is far above expectation, but not because those people seek each

other out. What attracts them to each other are the characteristics that make them poorly documented (few credits, minor roles, marginal periods, pseudonyms), and these are the same characteristics that make their gender hard to infer. It is data coverage disguised as social structure.

The reference measure therefore restricts the computation to edges where both endpoints have a determined gender: **80.3% of edges**. The difference is not cosmetic: **r goes from 0.0810 to 0.0432**, meaning that a third of the apparent homophily was an artifact.

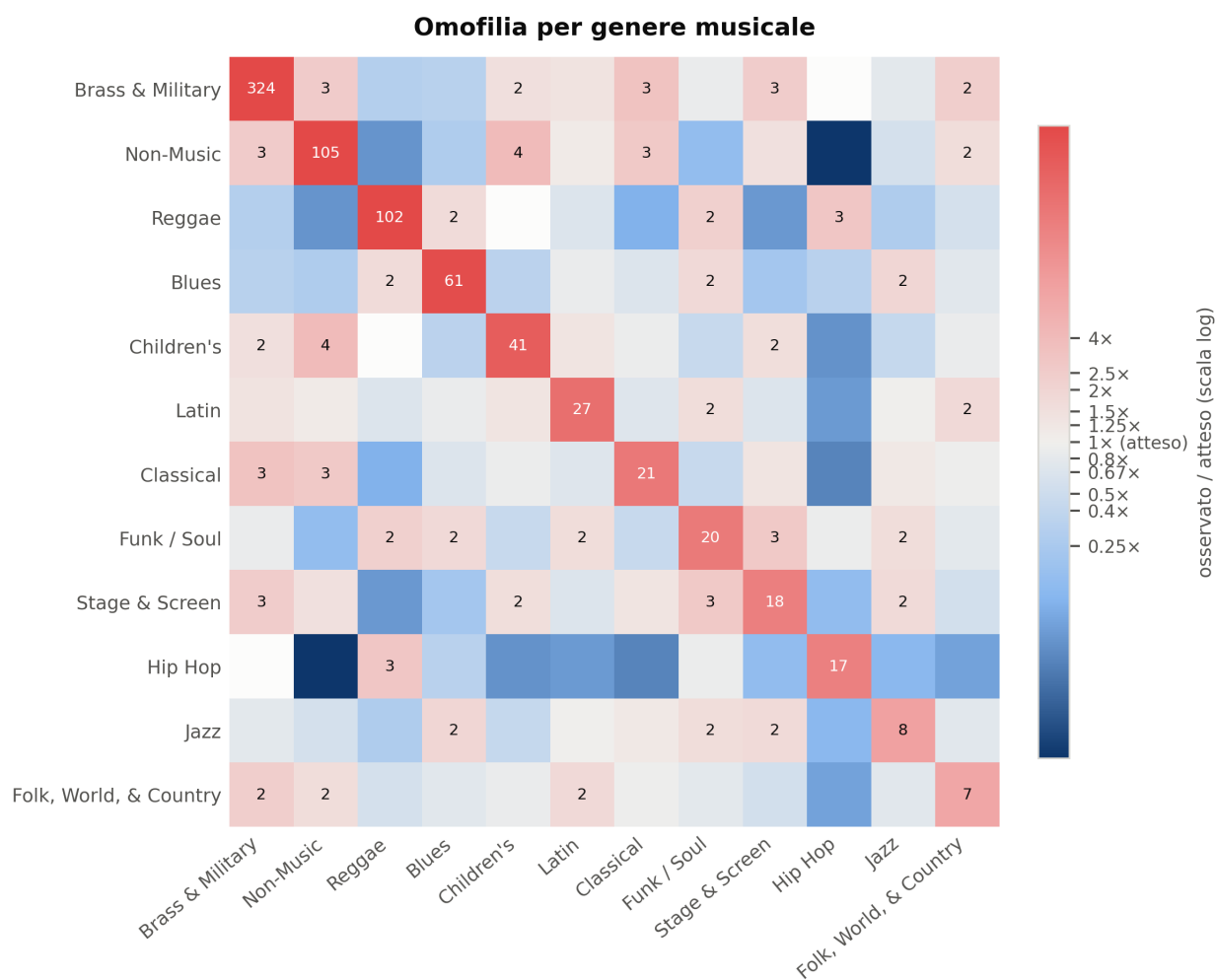
The same check was carried out on **musical genre**, where 'Unknown' is just as common, and it gives the **opposite** result: removing the undetermined cases, assortativity rises from 0.4970 to 0.5622. The reason is that artists without an assigned musical genre do not cluster together by genre (they have none), and so they dilute the diagonal instead of inflating it. Reporting both comparisons makes clear that excluding undetermined cases is a methodological choice applied uniformly, not an adjustment adopted where it was convenient: for gender it lowers the result, for musical genre it raises it.

The central result

Assortativity by **musical genre** is 0.497. This is a very high value: careers unfold within a genre, and collaborations follow genre boundaries almost as if they were industry boundaries.

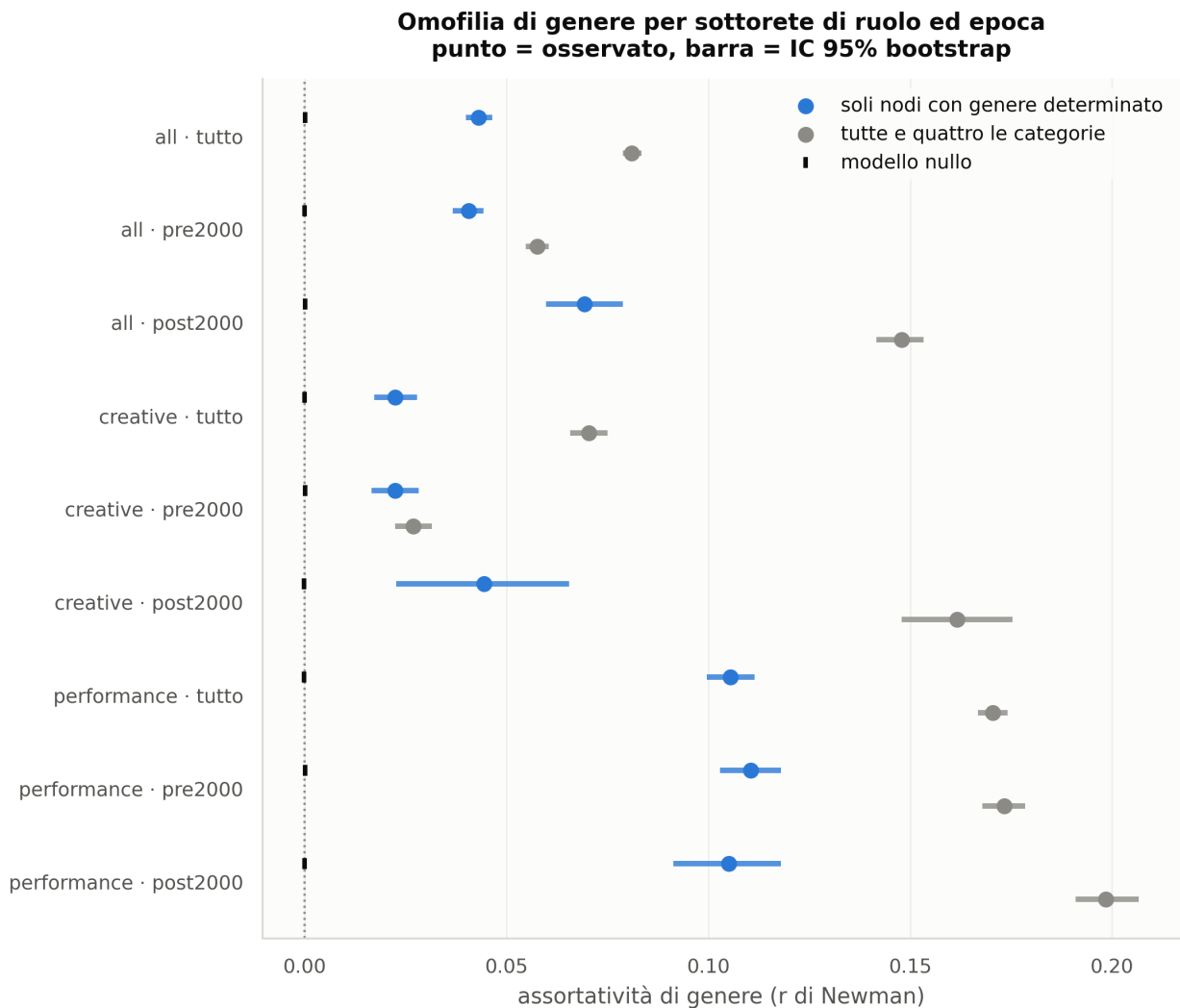
Assortativity by **gender** is 0.0432, about 12 times lower, with a confidence interval of 0.0399–0.0465 and a null-model value of 0.0001. Taken alone the number looks negligible; it is not, because with 586,040 edges even a small effect is measured with high precision and the interval lies entirely above zero.

The substantive reading is that **gender structures collaborations, but far less than musical specialization does**. Someone looking for a bass player looks within their own musical circle far more systematically than among people of their own sex. This does not make gender homophily irrelevant: it makes musical genre the main channel through which it operates, as the next paragraph shows.



Homophily by musical genre. The diagonal dominates the image. The most closed genres are not necessarily the largest: closure measures how much a scene recruits from within, not how populous it is. Off-diagonal cells above 1 indicate the pairs of genres between which there is a real flow of musicians: the permeable boundaries of the system.

5.2 Homophily over time: the counterintuitive result



Gender homophily by role subnetwork and period. The dot is the observed value, the bar the 95% bootstrap confidence interval, the vertical tick the null-model value. The expectation from the literature was a **loosening** after 2000. On nodes with determined gender only, the data say the opposite, but with a much smaller magnitude than the four-category measure shown in the figure suggests: 0.0406 versus 0.0693.

Before it can be interpreted, the result needs cleaning up. On the naive four-category measure the jump is spectacular, from 0.0576 to 0.1479: more than double. On nodes with determined gender only it shrinks to 0.0406 → 0.0693. The reason is that the share of usable edges drops sharply between the two periods, from 84.4% to 65.6%, because recent artists are on average less well documented: more unknown, hence more spurious apparent homophily.

What remains after the correction is nonetheless an increase, with intervals (0.0367–0.0443 versus 0.0598–0.0788) that do not overlap. The result holds, but it should be described for what it is: a moderate increase, not a doubling.

And the increase is not widespread: it comes entirely from one part of the network. Broken down by role, homophily in performance roles is essentially flat over

time (0.1106 before 2000, 0.1050 after), while homophily in creative roles **doubles**, from 0.0224 to 0.0444. Any explanation of the increase must therefore concern the way music is produced and written, not the way it is played.

Two readings remain possible, and the data at hand do not allow a definitive choice between them.

First reading: the increase is real. After 2000 the way music is produced changes: recording becomes decentralized, and large studios with mixed, fixed personnel give way to small projects built on personal networks. Personal networks mean more homophilous networks. In this reading the increase is not a cultural setback but the structural effect of a different production technology, and the fact that it concerns only creative roles, which are precisely those affected by the decentralization of studios, supports this explanation.

Second reading: the increase is compositional. Newman's r depends on the marginals. After 2000 there are more women, hence more opportunities for woman-woman ties; with the same propensity, a larger minority group mechanically produces a higher measured assortativity. The degree-preserving null model corrects for degree, **not** for this difference in composition between the periods.

The ERGM is needed precisely to settle this point, and **the verdict is in**: once the network's tendency to close triangles is controlled for, the difference in gender homophily between the two periods **is not statistically distinguishable from zero** (for women 0.709 versus 0.776, $p = 0.88$; for men 0.137 versus 0.080, $p = 0.60$).

What does increase, and very sharply, is **triadic closure**: the g_{wesp} coefficient goes from 2.217 to 2.892 ($p < 0.0001$). Italian recorded music after 2000 has not become more homophilous by gender: it has become more **closed into triangles**. People increasingly work within dense groups whose members already all know one another. In an environment where women are 14.8%, a more triangular structure mechanically produces more observed woman-woman ties, and hence a higher measured assortativity, without anyone having changed the criterion for choosing whom to work with.

This is the most interesting answer in the study, because it shifts the object: the problem is not a gender preference that has grown stronger, but a recruitment structure that has closed in on itself. These are two different things, also from the point of view of anyone wishing to intervene. The details of the test are in section 7.3.

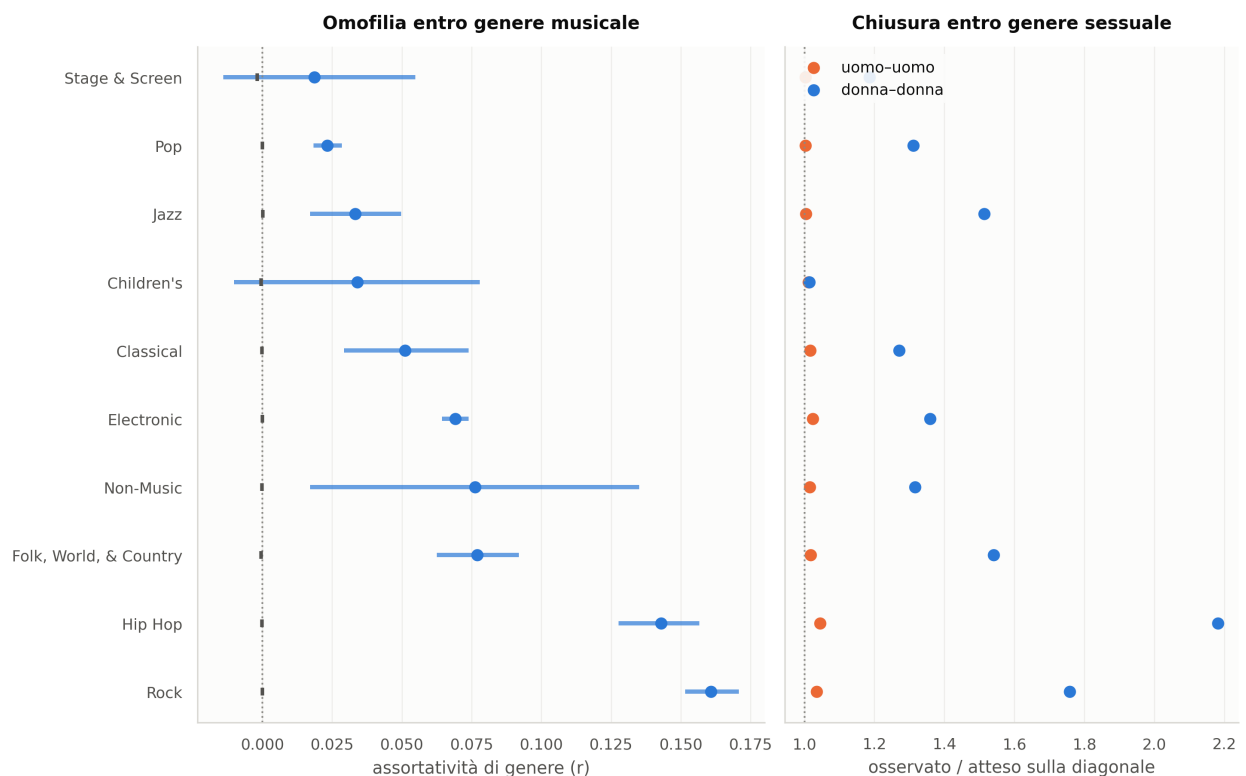
5.3 Producers and performers: two trades, two homophilies (RQ5)

The distinction between creative roles (production, writing, arrangement) and performance roles (vocals, instruments, conducting) is research question RQ5, and the answer is clear-cut: on nodes with determined gender only, homophily is **0.0224 in the creative network** versus **0.1054 in the performance network**.

Creative collaboration is therefore *less* segregated by gender than performance collaboration, by a factor of 4.7. This result has to be read together with the composition

figures: creative roles are by far the most male in the entire corpus, and precisely for this reason the homophily measured there is low, since where the majority is overwhelming there is almost no room to depart from chance. The segregation of creative roles shows up in **who gets in**, not in **who works with whom**; that of performance roles, where women are more present, also shows up in the structure of collaborations. These are two different forms of closure, and conflating them would lead to the conclusion that music production is the most open part of the system, which is the opposite of what the counts say.

5.4 Homophily within each musical genre



On the left, gender homophily computed separately within each musical genre, with the null model as reference. **On the right**, the decomposition of the diagonal: how far man-man ties and woman-woman ties exceed expectation. Reading the two panels together is the core of RQ2. An observed/expected value close to 1 for men and well above 1 for women describes a precise situation: men do not seek each other out more than chance would predict (they have no need to: they are the majority, and chance already puts them together), while women cluster together much more than expected. This is the form homophily takes when a minority operates within a majority: not symmetric segregation, but clustering of the minority group.

musical genre	edges	r gender	ci_lo	ci_hi	r_null	O/E M-M	O/E F- F	artists	share of women
Electronic	128,659	0.0692	0.0643	0.0739	0.0000	1.0241	1.3588	19,969	0.0985
Pop	109,924	0.0233	0.0183	0.0284	0.0001	1.0028	1.3115	9,446	0.1637
Rock	41,007	0.1608	0.1515	0.1708	0.0001	1.0353	1.7579	16,617	0.0806
Hip Hop	17,507	0.1429	0.1277	0.1567	-0.0001	1.0445	2.1820	1,822	0.0565
Folk, World, & Country	13,639	0.0769	0.0625	0.0919	-0.0005	1.0175	1.5411	4,863	0.1143
Jazz	11,762	0.0333	0.0171	0.0497	0.0001	1.0044	1.5144	3,124	0.1172
Classical	5,264	0.0512	0.0292	0.0739	-0.0002	1.0166	1.2704	3,674	0.2131
Stage & Screen	2,693	0.0186	-0.0141	0.0548	-0.0019	1.0035	1.1863	641	0.1700
Non-Music	1,106	0.0762	0.0171	0.1351	-0.0002	1.0151	1.3163	477	0.1698
Children's	1,075	0.0340	-0.0101	0.0780	-0.0005	1.0120	1.0144	676	0.3240

Table: Gender homophily within each musical genre (RQ2)

Full data in *tables/t3_omofilia_per_genere_musicale.csv* and *tables/t3_omofilia_per_genere_musicale.tex*.

This is the point on which the Italian data **contradict current expectations**. The literature usually reports stronger homophily *among men*. Here the woman-woman observed/expected ratio is systematically **higher** than the man-man ratio. The explanation is not that women are more closed: it is that with a female share of 14.8% the expected value for a woman-woman tie under randomness is very low, and a modest real clustering is enough to produce a high ratio. The man-man index, by contrast, is squeezed toward 1 because the majority cannot depart much from chance. **The two indices are not comparable as if they measured the same thing**, and it is the ERGM, which estimates a propensity rather than a ratio, that provides the correct comparison.

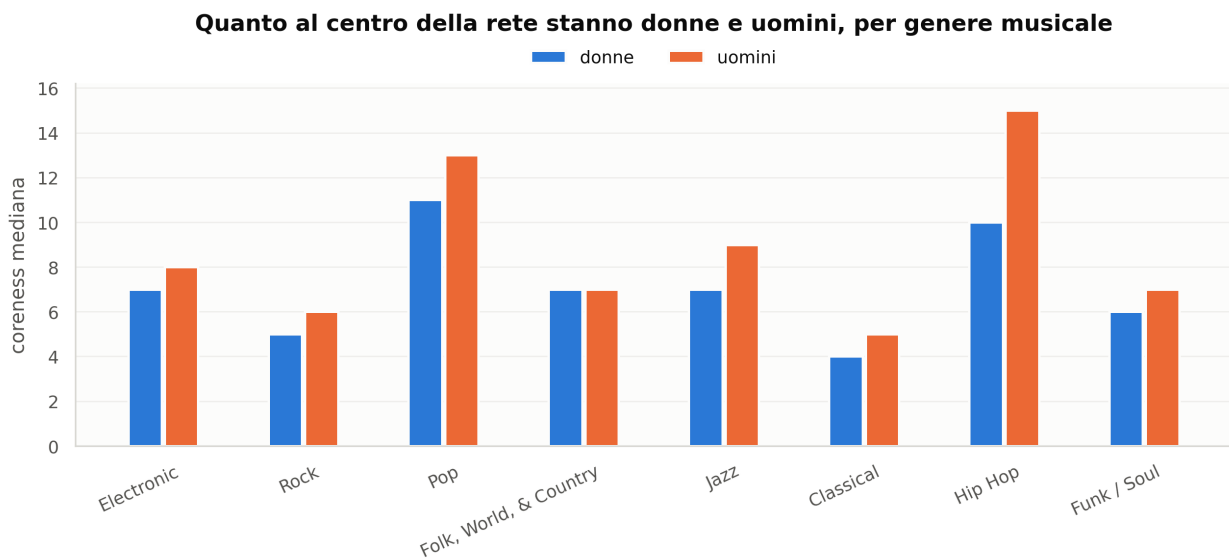
6. Are women at the center or at the margins? (RQ3)

The question that the literature calls the *Smurfette pattern* is whether women, where they are present, occupy structurally peripheral positions: present enough to be counted, never at the center.

gender	n	eigenvector (median)	eigenvector (mean)	betweenness (median)	coreness (median)	coreness (mean)	median degree	strength (median)	release (median)
F	6,961	0.000000	0.000055	0.000000	6.000000	9.011062	7.000000	5.190000	8.000000
M	44,869	0.000000	0.000174	0.000002	6.000000	10.028327	8.000000	6.490000	9.000000

Table: Position in the giant component by gender

Full data in `tables/t3_posizione_per_genere.csv` and `tables/t3_posizione_per_genere.tex`.



Median coreness by gender, within each musical genre. Coreness indicates which layer of the network's dense core a person belongs to: as a measure of belonging to the center it is more robust than degree centrality, because it is not inflated by having many occasional collaborators. The side-by-side bars allow a direct comparison within the same musical genre, which is the right comparison: comparing a pop singer with a jazz session musician would say nothing about gender and a great deal about the structure of the two scenes.

musical genre	gender	n	coreness (median)	eigenvector (median)
Blues	F	53	5.000000	0.000000
Blues	M	305	6.000000	0.000000
Brass & Military	F	2	7.500000	0.000000
Brass & Military	M	46	8.500000	0.000000
Children's	F	154	8.000000	0.000001
Children's	M	245	7.000000	0.000001
Classical	F	436	4.000000	0.000000
Classical	M	1,642	5.000000	0.000000
Electronic	F	1,722	7.000000	0.000000
Electronic	M	10,938	8.000000	0.000000
Folk, World, & Country	F	413	7.000000	0.000000
Folk, World, & Country	M	2,760	7.000000	0.000000
Funk / Soul	F	77	6.000000	0.000000
Funk / Soul	M	473	7.000000	0.000000
Hip Hop	F	72	10.000000	0.000000
Hip Hop	M	966	15.000000	0.000000
Jazz	F	245	7.000000	0.000000
Jazz	M	1,919	9.000000	0.000000
Latin	F	29	7.000000	0.000000
Latin	M	150	7.500000	0.000000
Non-Music	F	62	5.500000	0.000000
Non-Music	M	298	7.000000	0.000000
Pop	F	1,150	11.000000	0.000004
Pop	M	5,608	13.000000	0.000003
Reggae	F	14	4.500000	0.000000
Reggae	M	166	6.000000	0.000000
Rock	F	774	5.000000	0.000000
Rock	M	8,587	6.000000	0.000000
Stage & Screen	F	88	9.000000	0.000002
Stage & Screen	M	394	11.000000	0.000002

Table: Network position by gender within musical genre

Full data in `tables/t3_posizione_per_genere_musicale.csv` and `tables/t3_posizione_per_genere_musicale.tex`.

6.1 The comparison holding activity and cohort constant

Raw medians are not enough. Those who release more are more central, and the women in the population release less: the median number of releases is 8 for women versus 9 for men, and median coreness follows (6 versus 6). Without controlling for activity, one would be measuring the difference in how much people release and calling it a difference in position.

The regression therefore compares people with the same activity, the same debut cohort and the same musical genre.

term	coef	se	z	p	ci_lo	ci_hi	outcome
Intercept	0.8390	0.0966	8.6891	0.0000	0.6497	1.0282	eigenvector
C(gender)[T.F]	0.1801	0.1754	1.0265	0.3047	-0.1638	0.5239	eigenvector
C(genere)[T.Blues]	-0.0114	0.0764	-0.1498	0.8809	-0.1612	0.1383	eigenvector
C(genere)[T.Children's]	-0.1781	0.1203	-1.4802	0.1388	-0.4140	0.0577	eigenvector
C(genere)[T.Classical]	-0.6635	0.0683	-9.7184	0.0000	-0.7974	-0.5297	eigenvector
C(genere)[T.Electronic]	-0.1863	0.0627	-2.9721	0.0030	-0.3092	-0.0634	eigenvector
C(genere)[T.Folk, World, & Country]	-0.2480	0.0663	-3.7436	0.0002	-0.3779	-0.1182	eigenvector
C(genere)[T.Funk / Soul]	0.0681	0.0760	0.8962	0.3702	-0.0808	0.2171	eigenvector
C(genere)[T.Hip Hop]	-0.3351	0.0680	-4.9293	0.0000	-0.4683	-0.2018	eigenvector
C(genere)[T.Jazz]	-0.1980	0.0674	-2.9391	0.0033	-0.3300	-0.0660	eigenvector
C(genere)[T.Non-Music]	-1.4207	0.0969	-14.6663	0.0000	-1.6105	-1.2308	eigenvector
C(genere)[T.Pop]	1.0434	0.0662	15.7589	0.0000	0.9137	1.1732	eigenvector
C(genere)[T.Rock]	0.0077	0.0629	0.1223	0.9027	-0.1156	0.1310	eigenvector
C(genere)[T.Stage & Screen]	0.3258	0.0914	3.5651	0.0004	0.1467	0.5049	eigenvector
C(coorte)[T.1950]	0.2381	0.0906	2.6284	0.0086	0.0605	0.4156	eigenvector
C(coorte)[T.1960]	-0.1393	0.0793	-1.7560	0.0791	-0.2947	0.0162	eigenvector
C(coorte)[T.1970]	-1.2591	0.0757	-16.6331	0.0000	-1.4075	-1.1108	eigenvector
C(coorte)[T.1980]	-1.8311	0.0739	-24.7652	0.0000	-1.9760	-1.6862	eigenvector
C(coorte)[T.1990]	-2.0707	0.0735	-28.1849	0.0000	-2.2147	-1.9267	eigenvector
C(coorte)[T.2000]	-2.1104	0.0735	-28.6968	0.0000	-2.2546	-1.9663	eigenvector
C(coorte)[T.2010]	-2.0327	0.0737	-27.5984	0.0000	-2.1770	-1.8883	eigenvector
C(coorte)[T.2020]	-1.7953	0.0763	-23.5242	0.0000	-1.9449	-1.6457	eigenvector
C(gender)[T.F]:C(genere)[T.Blues]	0.0528	0.2128	0.2483	0.8039	-0.3643	0.4700	eigenvector
C(gender)[T.F]:C(genere)[T.Children's]	-0.1451	0.2392	-0.6068	0.5440	-0.6138	0.3236	eigenvector
C(gender)[T.F]:C(genere)[T.Classical]	0.0347	0.1850	0.1875	0.8513	-0.3279	0.3973	eigenvector
C(gender)[T.F]:C(genere)[T.Electronic]	-0.0192	0.1768	-0.1085	0.9136	-0.3657	0.3274	eigenvector
C(gender)[T.F]:C(genere)[T.Folk, World, & Country]	-0.1365	0.1863	-0.7323	0.4640	-0.5016	0.2287	eigenvector
C(gender)[T.F]:C(genere)[T.Funk / Soul]	0.2858	0.2295	1.2451	0.2131	-0.1641	0.7357	eigenvector

Table: Network position (eigenvector) by gender and musical genre, holding activity and cohort constant

First 28 of 35 rows shown; the full table is in `tables/t3_regression_eigenvector.csv` and `tables/t3_regression_eigenvector.tex`.

term	coef	se	z	p	ci_lo	ci_hi	outcome
Intercept	1.1580	0.0329	35.2227	0.0000	1.0936	1.2224	coreness
C(gender)[T.F]	0.0362	0.0792	0.4572	0.6475	-0.1190	0.1915	coreness
C(genere)[T.Blues]	0.0051	0.0360	0.1416	0.8874	-0.0654	0.0756	coreness
C(genere)[T.Children's]	-0.0135	0.0396	-0.3415	0.7328	-0.0912	0.0642	coreness
C(genere)[T.Classical]	-0.2867	0.0278	-10.3191	0.0000	-0.3411	-0.2322	coreness
C(genere)[T.Electronic]	0.0474	0.0256	1.8549	0.0636	-0.0027	0.0976	coreness
C(genere)[T.Folk, World, & Country]	0.0516	0.0268	1.9281	0.0538	-0.0009	0.1041	coreness
C(genere)[T.Funk / Soul]	0.0480	0.0333	1.4400	0.1499	-0.0173	0.1134	coreness
C(genere)[T.Hip Hop]	0.3801	0.0291	13.0492	0.0000	0.3230	0.4372	coreness
C(genere)[T.Jazz]	0.1173	0.0273	4.2999	0.0000	0.0638	0.1707	coreness
C(genere)[T.Non-Music]	-0.1804	0.0366	-4.9333	0.0000	-0.2521	-0.1088	coreness
C(genere)[T.Pop]	0.1794	0.0260	6.9001	0.0000	0.1284	0.2304	coreness
C(genere)[T.Rock]	-0.0319	0.0255	-1.2487	0.2118	-0.0820	0.0182	coreness
C(genere)[T.Stage & Screen]	0.1865	0.0345	5.3991	0.0000	0.1188	0.2542	coreness
C(coorte)[T.1950]	0.0563	0.0249	2.2643	0.0236	0.0076	0.1051	coreness
C(coorte)[T.1960]	-0.0144	0.0219	-0.6585	0.5102	-0.0574	0.0286	coreness
C(coorte)[T.1970]	-0.1829	0.0216	-8.4552	0.0000	-0.2254	-0.1405	coreness
C(coorte)[T.1980]	-0.1984	0.0213	-9.3131	0.0000	-0.2401	-0.1566	coreness
C(coorte)[T.1990]	-0.2171	0.0213	-10.2125	0.0000	-0.2587	-0.1754	coreness
C(coorte)[T.2000]	-0.2610	0.0215	-12.1340	0.0000	-0.3031	-0.2188	coreness
C(coorte)[T.2010]	-0.3656	0.0222	-16.5025	0.0000	-0.4091	-0.3222	coreness
C(coorte)[T.2020]	-0.3616	0.0273	-13.2565	0.0000	-0.4151	-0.3082	coreness
C(gender)[T.F]:C(genere)[T.Blues]	-0.0177	0.1034	-0.1715	0.8638	-0.2204	0.1850	coreness
C(gender)[T.F]:C(genere)[T.Children's]	-0.0037	0.0927	-0.0396	0.9684	-0.1853	0.1780	coreness
C(gender)[T.F]:C(genere)[T.Classical]	-0.0692	0.0840	-0.8244	0.4097	-0.2338	0.0954	coreness
C(gender)[T.F]:C(genere)[T.Electronic]	-0.0092	0.0801	-0.1147	0.9087	-0.1661	0.1478	coreness
C(gender)[T.F]:C(genere)[T.Folk, World, & Country]	-0.0633	0.0834	-0.7589	0.4479	-0.2268	0.1002	coreness
C(gender)[T.F]:C(genere)[T.Funk / Soul]	-0.0660	0.1054	-0.6263	0.5311	-0.2725	0.1405	coreness

Table: Core membership (coreness) by gender and musical genre, holding activity and cohort constant

First 28 of 35 rows shown; the full table is in `tables/t3_regression_coreness.csv` and `tables/t3_regression_coreness.tex`.

6.2 The answer to RQ3

The “Smurfette” pattern is not observed in these data. Holding releases, cohort and musical genre constant, the main effect of being a woman on network position is not distinguishable from zero on any of the three measures:

measure	coefficient	95% CI	p
eigenvector	0.180	-0.164 – 0.524	0.305
coreness	0.036	-0.119 – 0.191	0.648
betweenness	-0.293	-0.625 – 0.039	0.083

The interactions with musical genre do not change this either: of 12 interaction terms estimated, 0 are significant at the 5% level. In these data there is no scene in which being a woman systematically pushes a person toward the periphery of the network.

This result must be stated precisely, because it lends itself to two mistaken readings of opposite sign.

It does not mean that there is no inequality. Women are 14.8% of the population and release less: the median number of releases is 8 versus 9. The inequality exists and is large, but it shows up **in access and in the volume of activity**, not in structural position at equal activity. Those who get in and manage to work, work in comparable positions.

Nor does it mean that controlling for activity is neutral. The number of releases is not an exogenous variable: it is itself the outcome of access processes that may be segregated. Controlling for activity answers the question “for the same career, does position differ?” and not the question “do careers differ?”. The answer to the first is negative; the answer to the second (section 4) is clearly positive.

The most interesting finding is the divergence between the median and the mean of eigenvector centrality: the median is higher for women (7.7×10^{-9} versus 3.9×10^{-9}) while the mean is lower. This means that the typical woman in the network is at least as connected as the typical man, but that the **extreme hubs**, the very few nodes whose centrality is orders of magnitude higher, are almost all men. Positional inequality, where it exists, lies in the tail, not in the body of the distribution.

7. The probability of collaborating, all else being equal: the ERGM (RQ4)

7.1 Why it is needed and how it was set up

All the measures so far are descriptive: they say *that* ties are distributed in a certain way, not *why*. An exponential random graph model (ERGM) instead estimates the probability that a tie exists, combining in a single model gender homophily, musical-genre homophily, level of activity, cohort and the network's tendency to close triangles. This last point is decisive: without a triadic closure term (`gwesp`), any tendency to cluster is wrongly attributed to whatever attribute is being examined.

The ERGM cannot be estimated on the full network, and this should be stated plainly rather than glossed over. With 70,712 nodes the space of possible graphs makes MCMC sampling impractical. The analysis therefore proceeded, as the design anticipated, by subnetworks: the five most populous musical genres, the two periods, and a snowball sample of the overall network as a reference. **The estimates apply to the subnetworks on which they are computed**, and by construction do not extend to the entire population.

Nodes with unknown gender are excluded from the ERGM. Keeping them as a fourth category would have produced a spurious homophily term that measures the structure of data coverage (the poorly documented collaborate with the poorly documented) rather than the structure of collaborations.

A **hierarchy of three models** is estimated instead of a single specification, because the triadic closure term is exactly the one that can make estimation fail, and the rest of the analysis should not be lost along with it:

- **M0** edges + `nodematch(gender, diff)` + controls: no edge-dependence term. This is the reference model and can always be estimated.
- **M1** M0 + `gwesp(0.25; fixed)`: the specification anticipated by the design. The triadic closure term makes the model nearly degenerate on highly clustered networks, and it may fail to converge.
- **M2** the better of M1 and M0, with `nodemix(gender)` in place of `nodematch(gender, diff)`. The two parameterizations are redundant with each other, and including both in the same model would make it unidentified.

The control terms enter **only if the attribute varies** within the subnetwork. Within a subnetwork of a single musical genre, `nodematch('musical_genre')` is identical to edges: including it would produce an unidentified model, and in the first runs this was exactly what prevented convergence.

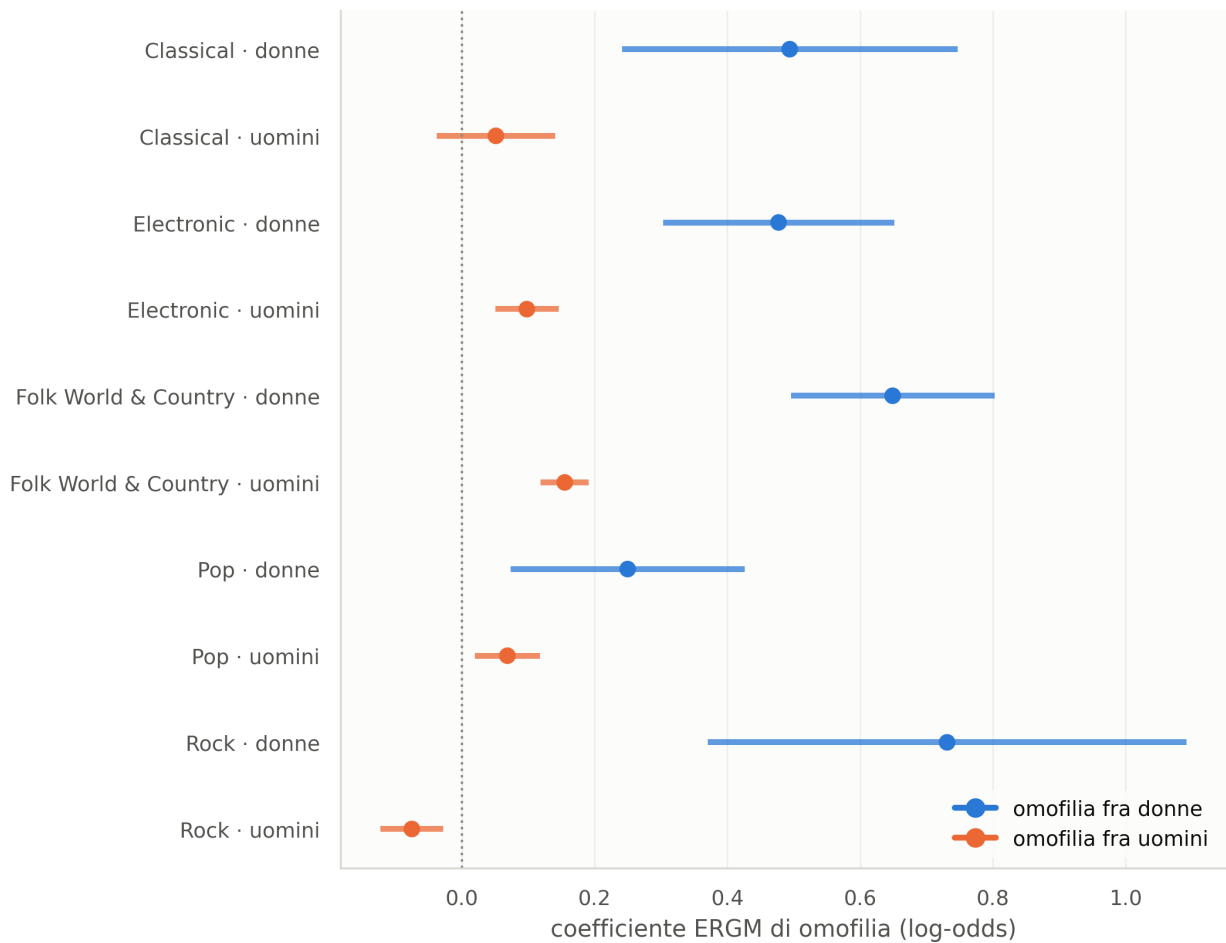
network	original nodes	estimated nodes	edges	sampled	converged	converged models	gwesp converg
genere_Electronic	10,499	1,255	4,968	True	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True
genere_Rock	4,938	1,438	4,972	True	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True
genere_Pop	5,236	1,057	4,989	True	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True
genere_Folk_World_&_Country	1,625	1,033	4,931	True	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True
genere_Classical	673	673	1,620	False	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True
epoca_pre2000	23,491	1,303	4,962	True	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True
epoca_post2000	9,400	1,373	4,961	True	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True
complessiva	36,388	722	4,982	True	True	M0_senza_gwesp, M1_con_gwesp, M2_nodemix_gwesp	True

Table: Summary of the ERGM estimates: size, sampling, convergence

Full data in `tables/t4_ergm_sintesi.csv` and `tables/t4_ergm_sintesi.tex`.

Subnetworks with at least one converged model: **8 of 8**; subnetworks in which the gwesp term also held: **8**. Where gwesp does not converge, the reported coefficients come from M0 and do **not** control for triadic closure: they should therefore be read as estimates that may attribute to homophily part of what is simply a tendency to form triangles. This is a limitation, and it is declared here rather than hidden behind a number.

Omofilia di genere a parità di attività, coorte e chiusura triadica nei cinque generi musicali più popolari



Gender homophily coefficients in the five most populous musical genres. Each coefficient is a log-odds: how much the probability of a tie increases (or decreases) if two artists share the same gender, *holding constant* activity, cohort, musical genre and triadic closure. This is the comparison that the descriptive measures in section 5.4 could not provide: here the male and female coefficients are on the same scale and directly comparable, because they measure a propensity and not an observed/expected ratio squeezed by the marginals. If the male coefficient exceeds the female one, the Italian pattern is in line with the literature (stronger homophily among men) and the inversion observed in section 5.4 was an artifact of women's rarity.

network	model	term	estimate	se	p	ci_lo	ci_hi
genere_Electronic	M0_senza_gwesp	edges	-9.6104	0.0947	-101.4612	-9.7961	-9.424
genere_Electronic	M0_senza_gwesp	nodematch.gender.F	0.6501	0.1016	6.3982	0.4509	0.8492
genere_Electronic	M0_senza_gwesp	nodematch.gender.M	0.1548	0.0343	4.5141	0.0876	0.2220
genere_Electronic	M0_senza_gwesp	nodematch.gender.mixed	1.7995	0.2002	8.9895	1.4072	2.1919
genere_Electronic	M0_senza_gwesp	nodecov.log_nrel	0.4587	0.0096	47.5998	0.4398	0.4776
genere_Electronic	M0_senza_gwesp	nodematch.cohort_decade	0.9364	0.0286	32.7475	0.8803	0.9924
genere_Electronic	M1_con_gwesp	edges	-8.3712	0.0990	-84.5544	-8.5653	-8.177
genere_Electronic	M1_con_gwesp	nodematch.gender.F	0.4771	0.0889	5.3685	0.3029	0.6513
genere_Electronic	M1_con_gwesp	nodematch.gender.M	0.0978	0.0244	4.0055	0.0499	0.1456
genere_Electronic	M1_con_gwesp	nodematch.gender.mixed	1.5036	0.1510	9.9574	1.2076	1.7996
genere_Electronic	M1_con_gwesp	nodecov.log_nrel	0.1270	0.0081	15.7610	0.1112	0.1428
genere_Electronic	M1_con_gwesp	nodematch.cohort_decade	0.5027	0.0332	15.1581	0.4377	0.5677
genere_Electronic	M1_con_gwesp	gwesp.fixed.0.25	2.3216	0.0627	37.0510	2.1988	2.4444
genere_Electronic	M2_nodemix_gwesp	edges	-7.5427	0.1086	-69.4677	-7.7556	-7.329
genere_Electronic	M2_nodemix_gwesp	mix.gender.F.M	-0.8361	0.0598	-13.9753	-0.9534	-0.718
genere_Electronic	M2_nodemix_gwesp	mix.gender.M.M	-0.6619	0.0618	-10.7108	-0.7830	-0.540
genere_Electronic	M2_nodemix_gwesp	mix.gender.F.mixed	-0.3314	0.1478	-2.2426	-0.6210	-0.041
genere_Electronic	M2_nodemix_gwesp	mix.gender.M.mixed	-0.7559	0.0811	-9.3256	-0.9148	-0.597
genere_Electronic	M2_nodemix_gwesp	mix.gender.mixed.mixed	0.5991	0.1927	3.1092	0.2214	0.9767
genere_Electronic	M2_nodemix_gwesp	nodecov.log_nrel	0.1282	0.0084	15.1873	0.1116	0.1447
genere_Electronic	M2_nodemix_gwesp	nodematch.cohort_decade	0.5610	0.0259	21.6217	0.5101	0.6118
genere_Electronic	M2_nodemix_gwesp	gwesp.fixed.0.25	2.1816	0.0703	31.0263	2.0437	2.3194
genere_Rock	M0_senza_gwesp	edges	-9.3291	0.0768	-121.5302	-9.4796	-9.178
genere_Rock	M0_senza_gwesp	nodematch.gender.F	0.8337	0.2548	3.2716	0.3342	1.3331
genere_Rock	M0_senza_gwesp	nodematch.gender.M	-0.1000	0.0402	-2.4897	-0.1788	-0.021
genere_Rock	M0_senza_gwesp	nodematch.gender.mixed	1.2413	0.2113	5.8744	0.8271	1.6554
genere_Rock	M0_senza_gwesp	nodecov.log_nrel	0.4589	0.0080	57.4936	0.4433	0.4746
genere_Rock	M0_senza_gwesp	nodematch.cohort_decade	1.3306	0.0286	46.5589	1.2746	1.3867
genere_Rock	M1_con_gwesp	edges	-10.3651	0.1100	-94.2404	-10.5806	-10.14
genere_Rock	M1_con_gwesp	nodematch.gender.F	0.7312	0.1842	3.9686	0.3701	1.0922
genere_Rock	M1_con_gwesp	nodematch.gender.M	-0.0756	0.0240	-3.1486	-0.1227	-0.028
genere_Rock	M1_con_gwesp	nodematch.gender.mixed	-0.5234	0.3615	-1.4479	-1.2319	0.1851
genere_Rock	M1_con_gwesp	nodecov.log_nrel	0.0913	0.0046	19.7145	0.0823	0.1004
genere_Rock	M1_con_gwesp	nodematch.cohort_decade	0.7956	0.0220	36.1783	0.7525	0.8387
genere_Rock	M1_con_gwesp	gwesp.fixed.0.25	4.4500	0.0733	60.7193	4.3064	4.5936

network	model	term	estimate	se	p	ci_lo	ci_hi
genere_Rock	M2_nodemix_gwesp	edges	-9.8962	0.3335	-29.6768	-10.5498	-9.242
genere_Rock	M2_nodemix_gwesp	mix.gender.F.M	-0.3735	0.3349	-1.1153	-1.0299	0.2829
genere_Rock	M2_nodemix_gwesp	mix.gender.M.M	-0.5840	0.3462	-1.6868	-1.2626	0.0946
genere_Rock	M2_nodemix_gwesp	mix.gender.F.mixed	-0.9523	0.3283	-2.9006	-1.5958	-0.308
genere_Rock	M2_nodemix_gwesp	mix.gender.M.mixed	-0.5631	0.3339	-1.6863	-1.2176	0.0914

Table: ERGM coefficients by subnetwork

First 40 of 185 rows shown; the full table is in `tables/t4_ergm_coefficienti.csv` and `tables/t4_ergm_coefficienti.tex`.

How to read these coefficients, and what they say

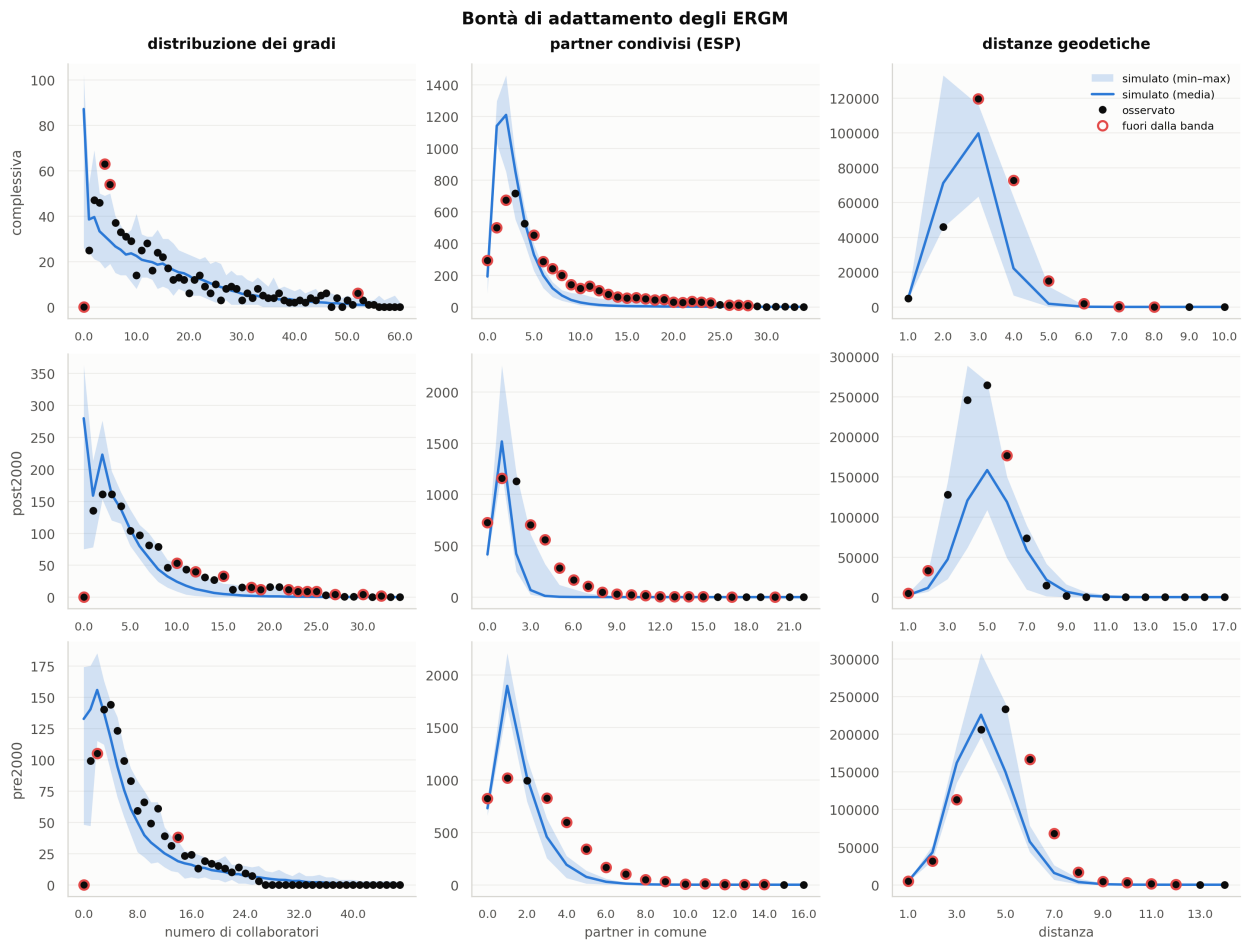
The coefficients are in log-odds; the `or` column is their exponential, that is, the factor by which the probability of a tie is multiplied. `nodematch.gender.F` says how much more likely a woman–woman pair is than a reference pair **holding everything else constant**: activity, cohort, musical genre and, decisively, the network’s tendency to close triangles.

The comparison between `nodematch.gender.F` and `nodematch.gender.M` is the result that section 5.4 could not provide. There, the observed/expected ratios were not comparable, because with a female share of 14.8% the expected value for a woman–woman tie is so low that any real clustering produces a large ratio. Here the problem does not arise: the two coefficients measure the same quantity on the same scale. **And the result holds: women’s propensity to work with women remains clearly higher than men’s propensity to work with men.** It is not an artifact of rarity; it is a property of the network.

In some subnetworks the male coefficient is **negative**: holding everything else constant, two men have a slightly *lower* probability of collaborating than the reference. This does not mean that men avoid each other. It means that, in an environment where they are the overwhelming majority, the edges term alone already produces almost all the male ties observed, and once that baseline is removed there is nothing left to attribute to a preference. This confirms the same asymmetry from the opposite side: where a group dominates, homophily has no way to show up as a measurable effect; where a group is rare, it shows up strongly.

The comparison between M0 and M1 also deserves attention. Moving from the model without triadic closure to the one with `gwesp`, the female homophily coefficient **decreases**: part of what looked like gender preference was in fact the network’s general tendency to close triangles (if A works with B and B with C, sooner or later A works with C, and in an environment where women are few, triangles among women form anyway). The `gwesp` term has a very large coefficient, which shows how strong that mechanism is. What remains after it is taken into account is genuine homophily.

7.2 Diagnostics: does the model actually describe this network?



Goodness of fit. For each subnetwork, three statistics of the observed network (points) are compared with those of the networks simulated from the estimated model (line and band): the degree distribution, the number of partners shared by each connected pair (ESP) and the geodesic distances. A model that captures the structure produces simulations whose band contains the observed values; where a point falls outside the band, the model is getting precisely that aspect wrong. ESP is the statistic to watch most closely, because it is the one the gwesp term should reproduce: if the observed values fall outside the band, triadic closure has not been captured and the homophily coefficients may have absorbed part of it. In the models estimated here the center of the distributions is reproduced well, but **the tail is not**: nodes with many collaborators and pairs with many shared partners are systematically more numerous than the model predicts (red circles). This is the known limitation of ERGMs on heavy-tailed networks, and it should be kept in mind when reading the coefficients: the model describes the typical musician well, and the few major collaborators less well.

Table *t4_ergm_mcmc* not available.

7.3 The pre/post-2000 test

model	term	estimate_pre	estimate_post	difference	se_diff	z	p
M0_senza_gwesp	edges	-10.1834	-10.1231	0.0603	0.1294	0.4657	0.64
M0_senza_gwesp	nodematch.gender.F	0.6129	0.3757	-0.2372	0.2690	-0.8817	0.37
M0_senza_gwesp	nodematch.gender.M	0.0546	0.1506	0.0960	0.0595	1.6145	0.10
M0_senza_gwesp	nodematch.gender.mixed	1.1725	1.8788	0.7063	1.0520	0.6714	0.50
M0_senza_gwesp	nodematch.musical_genre	1.8584	0.7809	-1.0775	0.0577	-18.6881	0.00
M0_senza_gwesp	nodecov.log_nrel	0.3983	0.4749	0.0766	0.0127	6.0327	0.00
M0_senza_gwesp	nodematch.cohort_decade	0.9962	0.8471	-0.1491	0.0465	-3.2085	0.00
M1_con_gwesp	edges	-8.9452	-8.7947	0.1506	0.1833	0.8216	0.41
M1_con_gwesp	nodematch.gender.F	0.7092	0.7763	0.0671	0.4368	0.1536	0.87
M1_con_gwesp	nodematch.gender.M	0.1368	0.0797	-0.0571	0.1091	-0.5234	0.60
M1_con_gwesp	nodematch.gender.mixed	1.2087	0.6167	-0.5920	1.3414	-0.4414	0.65
M1_con_gwesp	nodematch.musical_genre	0.9938	0.2883	-0.7055	0.0936	-7.5364	0.00
M1_con_gwesp	nodecov.log_nrel	0.1385	0.0803	-0.0582	0.0214	-2.7217	0.00
M1_con_gwesp	nodematch.cohort_decade	0.4768	0.3040	-0.1727	0.0844	-2.0462	0.04
M1_con_gwesp	gwesp.fixed.0.25	2.2169	2.8923	0.6754	0.0799	8.4490	0.00
M2_nodemix_gwesp	edges	-7.8019	-7.9375	-0.1357	0.3456	-0.3926	0.69
M2_nodemix_gwesp	mix.gender.F.M	-0.7275	-0.8898	-0.1623	0.3460	-0.4692	0.63
M2_nodemix_gwesp	mix.gender.M.M	-0.6808	-0.7937	-0.1129	0.3290	-0.3432	0.73
M2_nodemix_gwesp	mix.gender.F.mixed	-1.0455	0.0598	1.1053	0.6258	1.7661	0.07
M2_nodemix_gwesp	mix.gender.M.mixed	-1.0852	-0.9688	0.1164	0.4379	0.2659	0.79
M2_nodemix_gwesp	mix.gender.mixed.mixed	0.0206	0.2712	0.2506	3.2202	0.0778	0.93
M2_nodemix_gwesp	nodematch.musical_genre	1.0659	0.2194	-0.8465	0.0710	-11.9154	0.00
M2_nodemix_gwesp	nodecov.log_nrel	0.1052	0.0968	-0.0084	0.0179	-0.4684	0.63
M2_nodemix_gwesp	nodematch.cohort_decade	0.5239	0.3121	-0.2118	0.0677	-3.1283	0.00
M2_nodemix_gwesp	gwesp.fixed.0.25	2.1250	2.8679	0.7429	0.0908	8.1822	0.00

Table: Test of the difference between the pre-2000 and post-2000 ERGM coefficients

Full data in `tables/t4_ergm_differenza_epoche.csv` and `tables/t4_ergm_differenza_epoche.tex`.

This table is the formal test of the counterintuitive result of section 5.2, and its outcome is clear-cut.

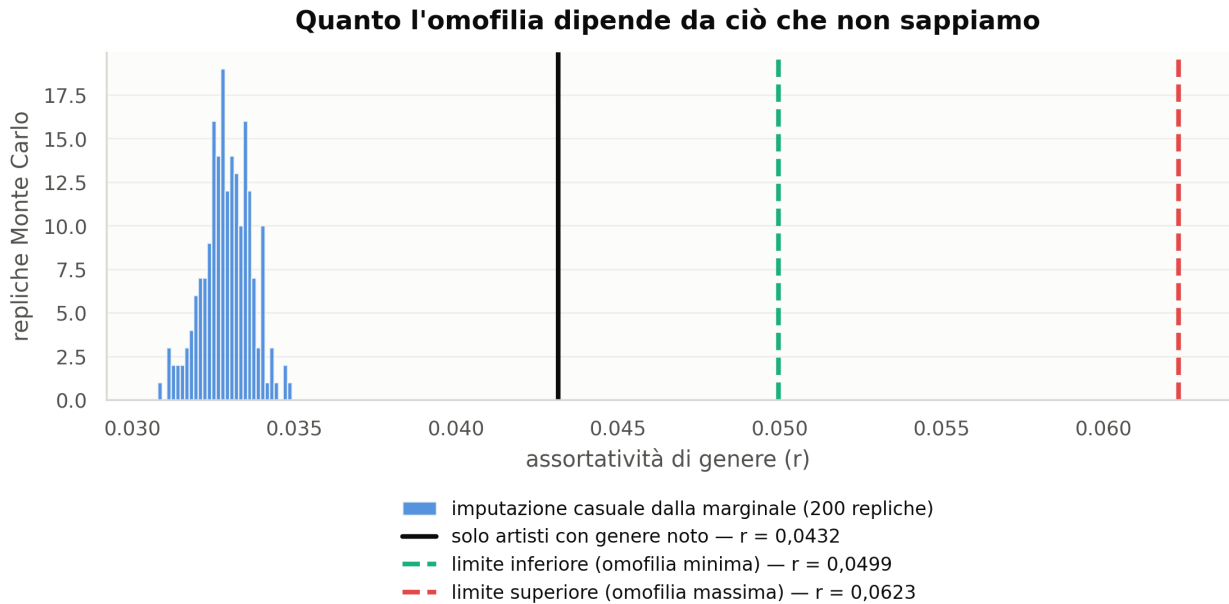
Gender homophily does not change between the two periods. For women the coefficient goes from 0.709 to 0.776 (difference 0.067, $p = 0.88$); for men from 0.137 to 0.080 (difference -0.057, $p = 0.60$). Neither difference comes close to significance.

Triadic closure, by contrast, changes a great deal. The $gwesp$ coefficient goes from 2.217 to 2.892, a difference of 0.675 with $p < 0.0001$: it is the only term in the model whose change between periods is statistically solid.

The joint reading is the one given in section 5.2: the increase in observed assortativity after 2000 is not a strengthening of gender preference, but the consequence of a network that closes into tighter groups. This is exactly the kind of confounding that a purely descriptive analysis cannot untangle, and the reason the ERGM was included in the design.

8. How well these results hold up

8.1 Uncertainty about gender



Monte Carlo distribution of gender assortativity as the imputation of the 22,136 artists without determined gender varies. The histogram is the distribution over 200 draws from the observed marginal. The two dashed lines at the sides are not estimates but **deliberately constructed bounds**: assigning each unknown artist the prevailing gender among their collaborators yields the maximum homophily compatible with the data; assigning the opposite gender yields the minimum. What matters is that **even the lower bound remains positive**: no assignment of the unknowns makes homophily disappear. The qualitative conclusion is robust; its exact magnitude is not.

scenario	count	mean	std	min	max	ci_lo	ci_hi
migliore_omofilia_min	1	0.0499	n/a	0.0499	0.0499	n/a	n/a
peggiore_omofilia_max	1	0.0623	n/a	0.0623	0.0623	n/a	n/a
solo_noti	1	0.0432	n/a	0.0432	0.0432	n/a	n/a
status_quo	200	0.0329	0.0007	0.0308	0.0349	0.0313	0.0343

Table: Gender assortativity under different imputation scenarios

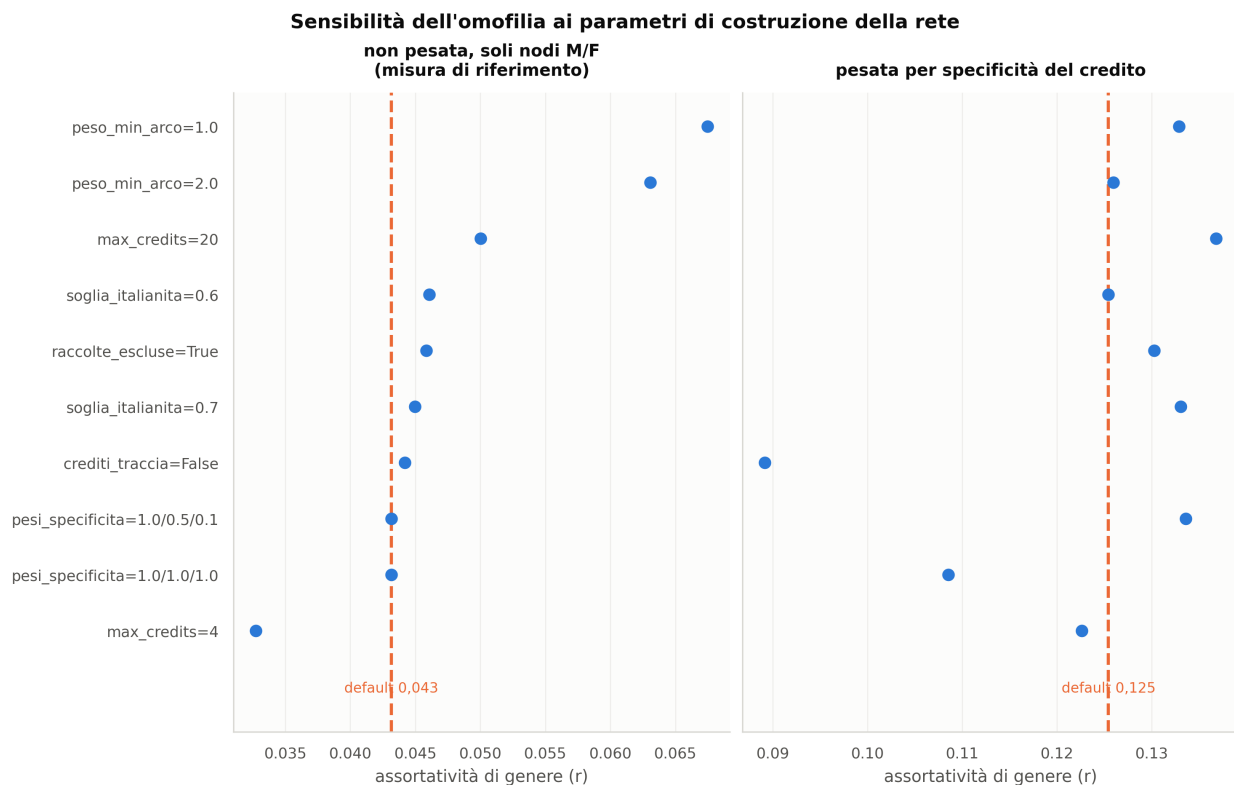
Full data in `tables/t5_montecarlo_genere.csv` and `tables/t5_montecarlo_genere.tex`.

The four numbers should be read together. On artists with determined gender only, assortativity is 0.0432. Imputing the unknowns by random draws from the observed marginal lowers it to 0.0329: random imputation can only dilute the structure, which is why the reference measure of this study is the first and not the second. The two neighborhood-based bounds, 0.0499 and 0.0623, delimit how large homophily could be if

the unknown artists systematically resembled their collaborators or systematically did not. **None of the four scenarios brings homophily to zero or below.**

It should be stated precisely what these bounds do not do: they concern only the unknown artists who have at least one collaborator of known gender. Those who collaborate only with other unknowns carry no usable information and are drawn from the marginal even in the extreme scenarios; assigning them by neighborhood majority would put them all in the same category, creating an artificial block that would inflate *both* extremes instead of bounding them.

8.2 Sensitivity to construction parameters



Each point is the gender assortativity obtained by changing a single parameter relative to the default configuration (dashed line). The axes varied are those that could reasonably change the result: the maximum number of artists a release can have for it to count as a collaboration, the minimum weight an edge must have, whether to exclude compilations, how strictly to define Italian identity, how to weight credit specificity, and whether or not to use track-level credits. The last is the most informative axis: switching off track credits means going back to a network built only on release-level co-presence, and it is the comparison that shows how much the track level actually adds.

variant	nod	edges	giant	r gender M/ F	r weighted	r musical genre
default	70,712	586,040	0.9538	0.0432	0.1254	0.4970
max_credits=4	50,042	204,330	0.9012	0.0327	0.1226	0.5278
max_credits=20	80,864	1,445,126	0.9702	0.0500	0.1368	0.4512
peso_min_arco=1.0	44,938	314,348	0.9133	0.0675	0.1329	0.5529
peso_min_arco=2.0	33,840	159,715	0.8569	0.0631	0.1260	0.5966
raccolte_escluse=True	65,549	508,524	0.9503	0.0459	0.1303	0.5037
crediti_traccia=False	69,806	546,310	0.9518	0.0442	0.0892	0.5006
pesi_specificita=1.0/1.0/1.0	70,712	586,040	0.9538	0.0432	0.1086	0.4970
pesi_specificita=1.0/0.5/0.1	70,712	586,040	0.9538	0.0432	0.1336	0.4970
soglia_italianita=0.6	63,488	500,418	0.9527	0.0461	0.1254	0.4875
soglia_italianita=0.7	53,916	390,999	0.9507	0.0450	0.1331	0.4795

Table: Sensitivity of key metrics to the network construction parameters

Full data in `tables/t5_sensibilita.csv` and `tables/t5_sensibilita.tex`.

Across the 10 variants tried, gender assortativity on determined nodes only stays between **0.0327 and 0.0675**, always positive and always of the same order of magnitude as the reference value (0.0432). No defensible choice of network construction reverses the conclusion.

What track-level credits actually add

The **weighted** assortativity column is the only one in which the hierarchy of credit specificity can show up, because changing the weights does not change which pairs of artists are connected: it changes how much they count. The comparison is instructive.

- with track credits and the default hierarchy: **0.1254**
- without track credits, that is, going back to release-level co-presence only: **0.0892**
- with all credits at the same weight: 0.1086
- with a steeper hierarchy (1 / 0.5 / 0.1): 0.1336

Switching off the track level lowers measured homophily by about 29%, and flattening the weights lowers it almost as much. The reading is that **the collaborations documented most specifically are also the most homophilous**: when two names appear together on the same track, and not just generically on the same record, the probability that they share the same gender is higher. A network built on release-level co-presence alone therefore underestimates segregation, because it mixes genuine collaboration with editorial cohabitation. This is the empirical justification for the design choice described in section 3.1.

8.3 Artists with a weak musical-genre assignment

variant	nodi	edges	density	giant share	r gender	r gender M/F	r gender weighted	r musical genre	share of women	median weight	ar
genre_top1	70,712	586,040	0.0002	0.9538	0.0810	0.0432	0.1254	0.4970	0.1477	1.0000	87
genre_top2	70,712	586,040	0.0002	0.9538	0.0810	0.0432	0.1254	0.4538	0.1477	1.0000	87
genre_exclude	45,704	349,998	0.0003	0.9505	0.0822	0.0486	0.1234	0.7045	0.1489	1.0000	56

Table: Key metrics using the main tag, the second tag, or excluding the artists with a weak assignment

Full data in `tables/t5_genere_debole.csv` and `tables/t5_genere_debole.tex`.

Artists whose main genre tag covers less than 40% of their releases are flagged `genre_weak`: there are 30,724 of them, that is, 35.2% of the population. The table compares three treatments (keeping them with the main tag, replacing it with the second tag, excluding them altogether) to show how far the conclusions on RQ2 depend on a musical-genre assignment that is uncertain by construction.

Technical appendix

A.1 Environment

- System: Linux-5.15.0-185-generic-x86_64-with-glibc2.35
- Python 3.9.12
- PostgreSQL 18.4 (database discogs, data on a spinning disk)
- R for the ERGM: installed in user space via micromamba (conda-forge), env opt/mamba/envs/ergm
- Global random seed: **20260920**
- Run date: 26 September 2026

package	version
gender_guesser	installed
matplotlib	3.8.1
networkx	3.2.1
numpy	1.24.2
pandas	1.5.3
psycpg2	2.9.10 (dt dec pq3 ext lo64)
pyarrow	18.1.0
scipy	1.11.3
seaborn	0.13.2
statsmodels	0.14.6

A.2 A note on performance that shaped the design

The PostgreSQL cluster resides on a **spinning** disk but was configured with `random_page_cost = 1.1`, a value tuned for solid-state drives. With that cost the planner prefers random-access paths that on a mechanical disk degrade to a few megabytes per second: the first version of the extraction ran at about 5 MB/s. The extraction sessions therefore set `random_page_cost = 4` and reduce parallelism, favoring sequential scans, and the Italian-share counts were rewritten as **a single aggregated pass** over `release_artist` instead of two repeated joins. These are session-level GUCs: they modify neither the server configuration nor the data.

Similarly, the computation of the mixing matrix was rewritten from `numpy.add.at` to `numpy.bincount` on flattened indices (numerically identical result, verified, about **50 times faster**), because without that rewrite the thousands of bootstrap and null-model replicates required by the design would not have been feasible.

A.3 Main queries

All queries are in `src/phase1_extract.py`. The most important one defines the population, in a single pass:

```
SELECT ra.artist_id,  
       count(*) FILTER (WHERE it.id IS NOT NULL) AS n_it,  
       count(*)                               AS n_all  
FROM release_artist ra  
LEFT JOIN (SELECT id FROM release WHERE country = 'Italy') it  
ON it.id = ra.release_id  
GROUP BY 1  
HAVING count(*) FILTER (WHERE it.id IS NOT NULL) >= 2;
```

The final selection (share, minimum number of releases, exclusions based on the artist records) then takes place on the data already saved locally, so that Phase 5 can vary the thresholds without rereading the database.

A.4 Run times

step	seconds
ERGM gwesp025_gwdeg	45,780.5
ERGM 2020s	13,466.7
ERGM 1950s	10,928.0
ERGM decay_alto	8,509.8
logit esatto su tutte le diadi	8,014.7
serie temporale esatta	5,449.4
betweenness	5,238.8
ERGM 1960s	4,602.8
ERGM gwesp050_stocapp	4,406.4
ERGM gwesp_esp0	3,585.5
rete intera: 20 riproiezioni randomizzate	3,342.4
ERGM genere_Rock	3,183.3
sensibilita'	2,578.6
ERGM 1940s	2,468.0
ERGM due_scale	1,687.4
ERGM genere_Folk_World_&_Country	1,643.7
ERGM epoca_post2000	1,612.0
ERGM complessiva	1,573.4
ERGM genere_Electronic	1,485.7
passata 1b (ricerca di linea)	1,460.8

A.5 Reproducibility

```
cd /media/disk2/datascience/analysis/gender_collab
./run_all.sh           # full run
./run_all.sh --from 3  # restart from Phase 3
./run_all.sh --force   # ignore checkpoints and recompute everything
```

Each phase writes a checkpoint in `data/*.parquet` and is skipped if the checkpoint exists. The raw extracted data are in `data/raw/`, the figures in `report/figures/`, and the tables in `report/tables/`, in both CSV and LaTeX.

A.6 Limitations, in order of severity

1. **Gender is inferred, and the manual validation has not been carried out.** The stratified sample and the scoring script are ready; without the validation, the measurement error of the inference remains unquantified.
2. **Italian identity is approximated by the country of release**, because `release_label` is empty. This conflates “Italian artist” with “artist released in Italy”.
3. **26.9% of the population has no musical genre**, because `release_genre` is empty and masters cover only part of the releases.
4. **No cross-check between sources:** iTunes was excluded by choice.
5. **Discogs is not a census.** It overrepresents vinyl, electronic music and collecting.
6. **The ERGM applies to the estimated subnetworks**, not to the whole network.
7. **Wikidata covers 6.0% of the population;** the name-based recovery step was not completed because the SPARQL service was repeatedly unavailable, responding with 429, 502 and 504 errors.